PARAMETRIC CAD · EXAMPLE COLLECTION

Everyday models Catalogue

19 parametric models from `tests/models/everyday/`: household, desk, workshop and typical machine parts – from the mug to the pillow block bearing.

The collection serves two purposes: as a template for your own parts (every model is a module with default values, all dimensions in mm) and as a test suite that uses the language broadly – grids and anchors, fillet and chamfer via zones, `shell`, `loft`, `sweep`, profiles, `include`/`use`, named instances as parts. The regression `tests/everyday/check.sh` checks all models; bugs and weaknesses found are listed in `tests/models/everyday/README.md`.

For each model the catalogue shows a page with description, rendering, views, key data, parameters and the language features used; assemblies have a second page with parts, measured distances and a section or the anchors. All values are measured with the CLI.

OPENING A MODEL

`mcad tests/models/everyday/mug.mcad`	desktop UI: editor and 3D
`mcad tests/models/everyday/housing/`	assembly as project (folder)
`mcad render tests/models/everyday/mug.mcad -o mug.png --views all`	headless: image
`mcad outline tests/models/everyday/mug.mcad`	parameters, grids, anchors, parts
`mcad measure tests/models/everyday/mug.mcad --volume --colors`	volume, mass per colour
`mcad export tests/models/everyday/gear.mcad --all-parts -o parts/`	STL/STEP/3MF per part

19
models

7
with multiple parts

36
parts

1.46 kg
PLA for everything

As of 2026-10-26, commit `83fbe92` · generated with `node tools/report/katalog.mjs` · images `cad render --view iso --perspective`, fitted · mass: PLA 1.24 g/cm³ · kernel time: `scene.stats.kernel_ms` from `cad eval --no-daemon`

Contents

NO.	MODEL, FILE	PARTS	BOUNDING BOX W × D × H MM	PLA	KERNEL	PAGE
01	Phone stand phone_stand.mcad	1	81.7 × 70.0 × 91.1	136 g	0.20 s	4
02	Mug with handle mug.mcad	1	116.6 × 81.9 × 95.0	96 g	1.52 s	5
03	Cable clip for the desk edge cable_clip.mcad	1	31.5 × 52.0 × 36.4	30 g	0.68 s	6
04	Wall hook wall_hook.mcad	1	32.0 × 51.0 × 80.0	23 g	0.75 s	7
05	Soap dish soap_dish.mcad	1	115.0 × 80.0 × 27.0	46 g	1.56 s	8
06	Pen box pen_box.mcad	1	99.6 × 70.0 × 95.0	143 g	1.20 s	9
07	Door stop door_stop.mcad	1	120.0 × 43.6 × 32.0	80 g	0.29 s	10
08	Pot trivet trivet.mcad	1	180.0 × 180.0 × 8.0	103 g	5.36 s	11
09	Flange plate flange.mcad	1	120.0 × 120.0 × 28.0	186 g	1.01 s	12
10	Pillow block 608 pillow_block.mcad	1	80.0 × 30.0 × 50.0	43 g	0.82 s	13
11	Mounting bracket with rib rib_bracket.mcad	1	60.0 × 40.0 × 50.0	28 g	0.58 s	14
12	Heat sink heat_sink.mcad	1	60.0 × 50.0 × 25.0	43 g	1.05 s	15
13	Spur gear pair Assembly gear.mcad	2	67.6 × 44.0 × 16.0	21 g	4.97 s	16
14	Pipe clamp Assembly pipe_clamp.mcad	3	60.0 × 64.0 × 52.0	32 g	1.79 s	18
15	Raspberry Pi mounting plate Assembly pi_plate.mcad	2	100.0 × 72.0 × 26.6	57 g	1.85 s	20
16	Electronics enclosure Assembly housing/housing.mcad	6	80.0 × 60.0 × 36.7	52 g	3.67 s	22
17	Hinge Assembly hinge/hinge.mcad	3	61.0 × 58.0 × 8.0	13 g	2.27 s	24
18	Plant pot with saucer Assembly planter/planter.mcad	2	113.7 × 113.7 × 96.4	133 g	1.15 s	26
19	Key rack Assembly key_rack/key_rack.mcad	6	200.0 × 40.0 × 70.0	196 g	3.41 s	28

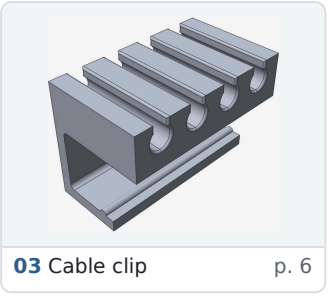
Overview



01 Phone stand p. 4



02 Mug with handle p. 5



03 Cable clip p. 6



04 Wall hook p. 7



05 Soap dish p. 8



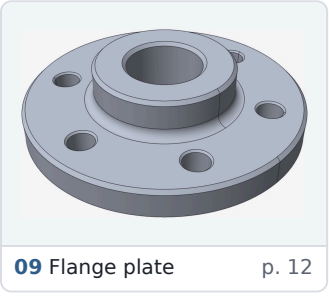
06 Pen box p. 9



07 Door stop p. 10



08 Pot trivet p. 11



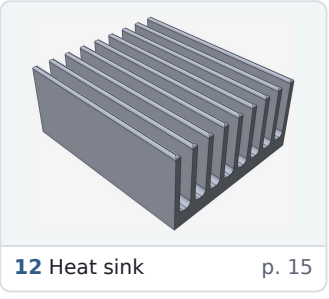
09 Flange plate p. 12



10 Pillow block 608 p. 13



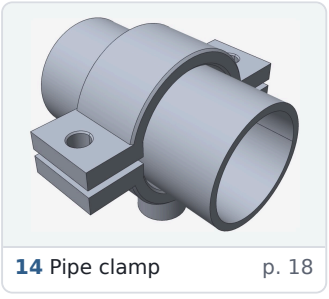
11 Mounting bracket p. 14



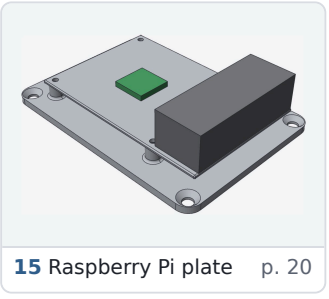
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13 Spur gear pair p. 16



14 Pipe clamp p. 18



15 Raspberry Pi plate p. 20



16 Electronics enclo… p. 22



17 Hinge p. 24



18 Plant pot p. 26

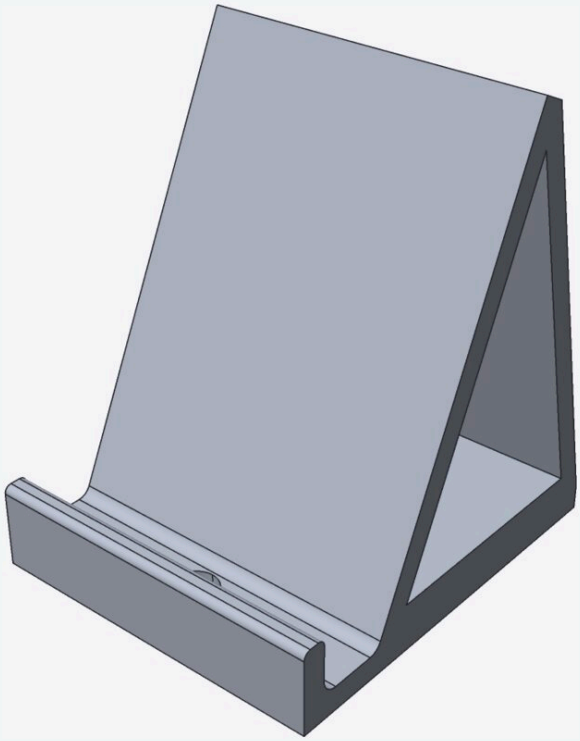


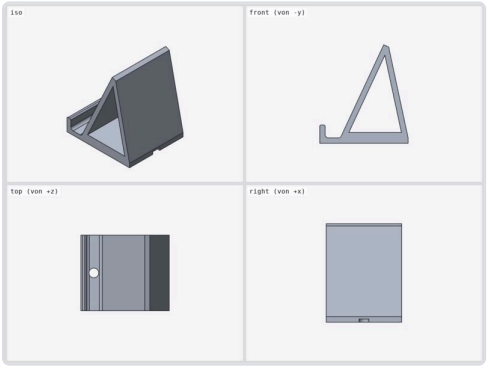
19 Key rack p. 28

01 Phone stand

Single part

Stand for a phone up to 12 mm thick. The side profile is a polygon of points of a 2D grid and is extruded over the width w = 70 mm; the backrest (95 mm, wall t = 5 mm) is at an angle a = 65° to the horizontal, a triangular window lightens it. A cable hole Ø 9 mm with a channel under the base takes the charging cable. Inner corners of lip and shelf as well as the top edge of the lip are filleted with two radii via zones on grid lines.





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	81.7 × 70.0 × 91.1 mm
Volume	110.0 cm³
Mass PLA (1.24 g/cm³)	136.4 g
Parts	1: stand
Colours / filaments	single colour (filament slot 1)
Kernel time (no cache)	0.20 s (20 nodes)

PARAMETERS (DEFAULT VALUES)

phone_stand() `w = 70` `a = 65` `backrest = 95` `t = 5` `lip = 12` `shelf = 14` `cable_d = 9`

LANGUAGE FEATURES

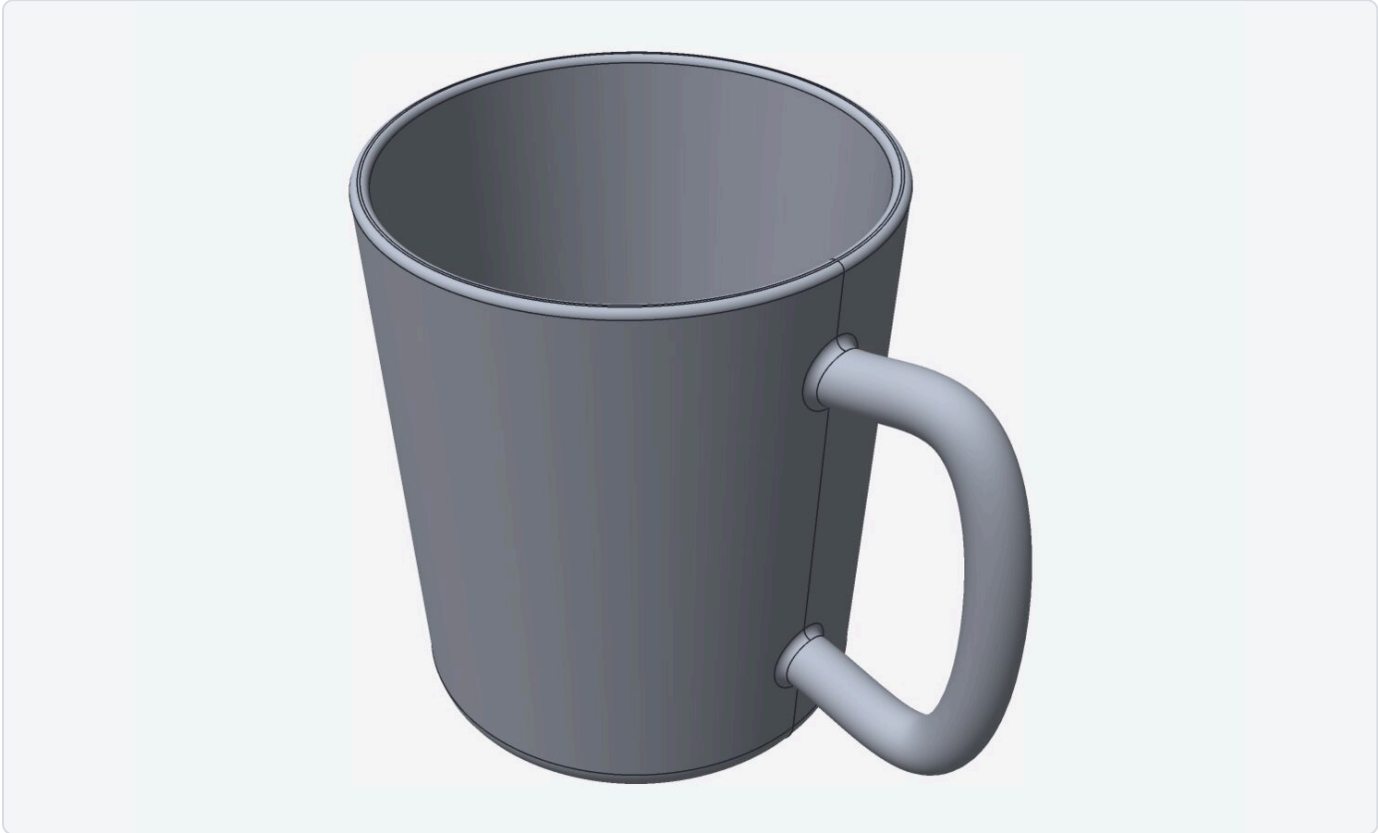
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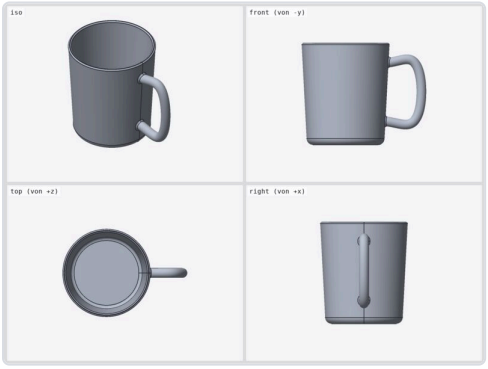
```
OPEN      cad tests/models/everyday/phone_stand.mcad
HEADLESS  cad render tests/models/everyday/phone_stand.mcad -o bild.png --views all
```

02 Mug with handle

Single part

Drinking mug as a conical solid of revolution (radius at the bottom r_bottom = 36 mm, at the top r_top = 41 mm, height h = 95 mm), opened at the top with shell and a 2.8 mm wall. The bottom edge is filleted with r 6 mm via a zone on the bottom plane, the drinking rim via a second zone. The handle (Ø 9 mm) is a sweep of a circle along a spline path whose ends sit radially in the wall, attached with union(fillet = …).





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	116.6 × 81.9 × 95.0 mm
Volume	77.7 cm³
Mass PLA (1.24 g/cm³)	96.4 g
Parts	1: b
Colours / filaments	single colour (filament slot 1)
Kernel time (no cache)	1.52 s (13 nodes)

PARAMETERS (DEFAULT VALUES)

mug() `r_bottom = 36` `r_top = 41` `h = 95` `wall = 2.8` `floor_r = 6` `handle_d = 9`

LANGUAGE FEATURES

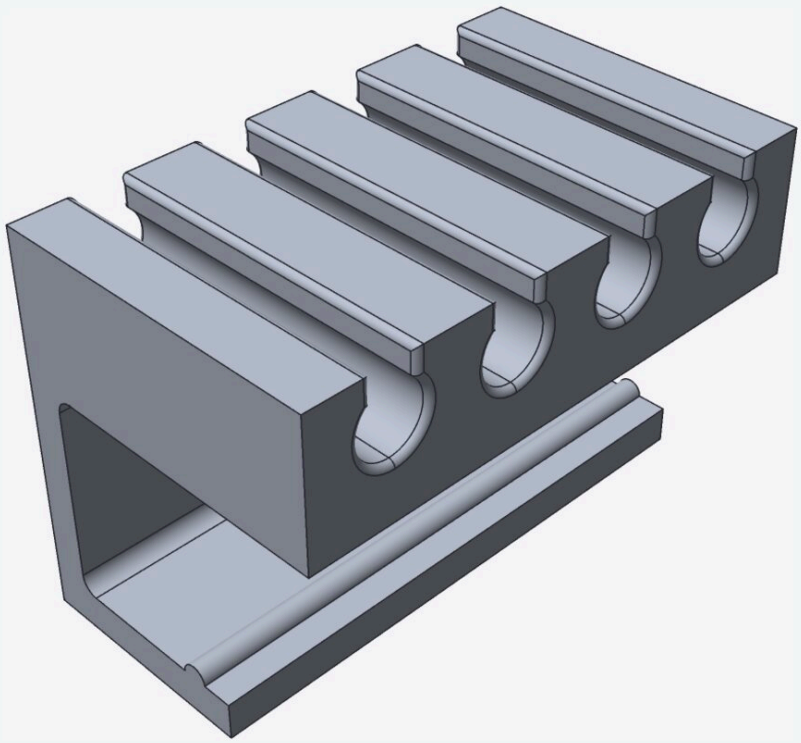
grid · anchor · anchor with dir · named instances · function · polygon · spline · rotate_extrude · sweep · shell · fillet zones · zone · union(fillet)

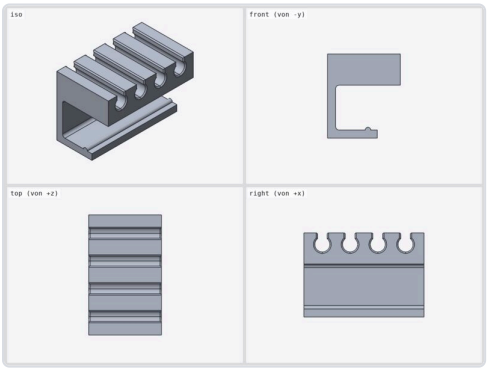
```
OPEN      cad tests/models/everyday/mug.mcad
HEADLESS  cad render tests/models/everyday/mug.mcad -o bild.png --views all
```

03 Cable clip for the desk edge

Single part

Cable holder that clips onto a desk top of thickness = 19 mm (clearance 0.4 mm). The C clamp is an extruded profile (wall 3.5 mm, arms 28 and 18 mm) with a clamping bump; above it sits a block with n = 4 cable channels Ø 6 mm at a pitch of 12 mm, inserted from the top. The channels lie on indexed grid planes, their edges are filleted with difference(fillet = …), the inner corners of the clamp via zones.





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	31.5 × 52.0 × 36.4 mm
Volume	24.5 cm³
Mass PLA (1.24 g/cm³)	30.3 g
Parts	1: `clip`
Colours / filaments	single colour (filament slot 1)
Kernel time (no cache)	0.68 s (28 nodes)

PARAMETERS (DEFAULT VALUES)

cable_clip() `thickness = 19` `clearance = 0.4` `t = 3.5` `arm_top = 28` `arm_bottom = 18` `n = 4` `cable_d = 6` `pitch = 12` `block_h = 10`

LANGUAGE FEATURES

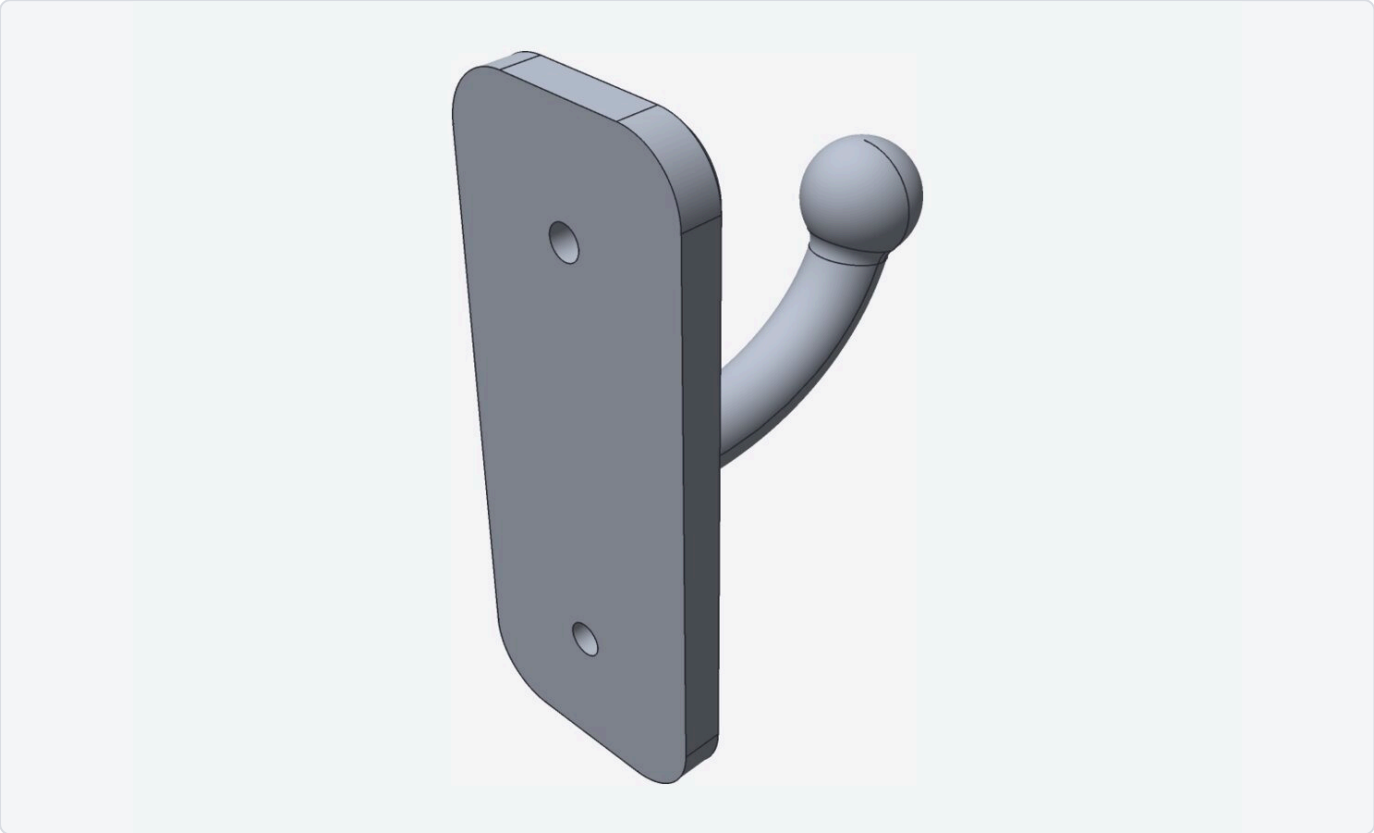
grid · indexed grid planes · anchor · anchor with dir · named instances · list comprehension · for · polygon · linear_extrude · fillet zones · zone · difference(fillet/chamfer) · box(a, b) · cylinder(a, b, r)

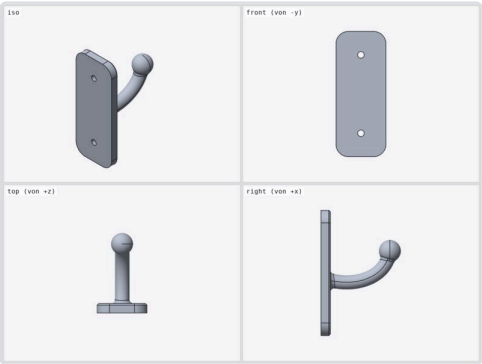
```
OPEN      cad tests/models/everyday/cable_clip.mcad
HEADLESS  cad render tests/models/everyday/cable_clip.mcad -o bild.png --views all
```

04 Wall hook

Single part

Hook for a jacket or towel: wall plate 32 × 80 × 6 mm with filleted corners (r 8 mm), chamfered front edge and two countersunk holes (Ø 4.2 mm, head Ø 8.5 mm). Chamfer and fillet are nested, so that the chamfer also covers the boundary edges of the fillet. The hook (Ø 9 mm, reach 38 mm) is a sweep along a spline path with a ball knob Ø 14 mm, attached with union(fillet = ...).





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	32.0 × 51.0 × 80.0 mm
Volume	18.8 cm³
Mass PLA (1.24 g/cm³)	23.3 g
Parts	1: hook
Colours / filaments	single colour (filament slot 1)
Kernel time (no cache)	0.75 s (31 nodes)

PARAMETERS (DEFAULT VALUES)

wall_hook() `b = 32` `h = 80` `t = 6` `corner = 8` `screw_d = 4.2` `head_d = 8.5` `hook_d = 9` `reach = 38` `knob_d = 14`

LANGUAGE FEATURES

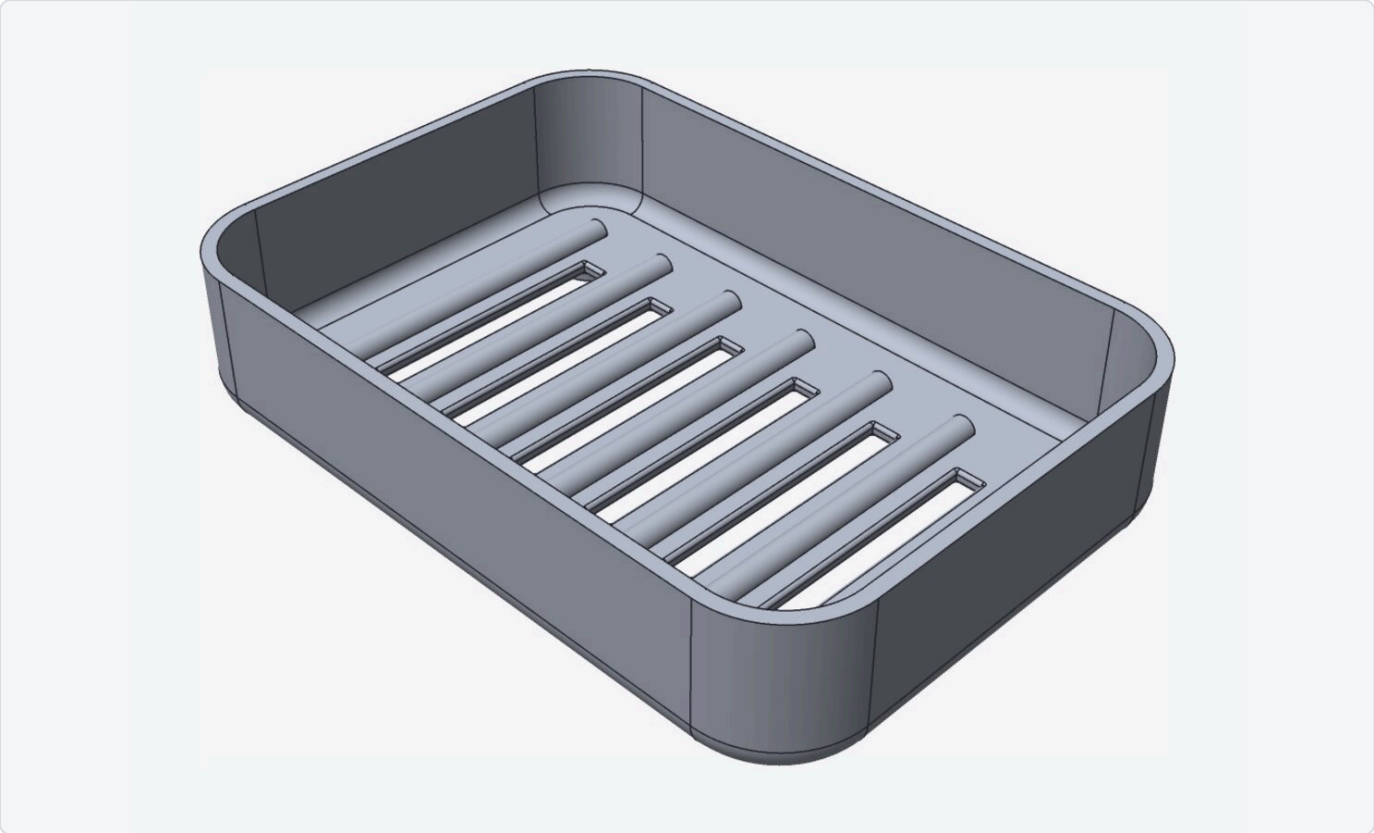
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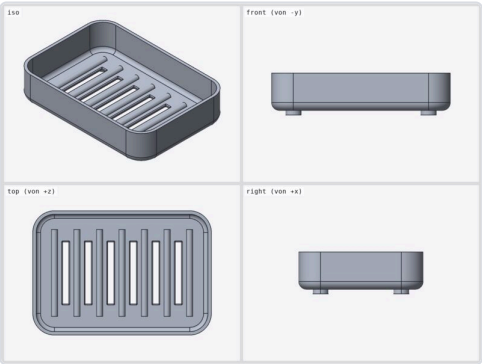
```
OPEN      cad tests/models/everyday/wall_hook.mcad
HEADLESS  cad render tests/models/everyday/wall_hook.mcad -o bild.png --views all
```

05 Soap dish

Single part

Soap dish 115 × 80 × 24 mm with drainage. The tray is a box whose vertical corners (r 14 mm) and bottom edges are filleted via nested zones and which shell (wall 2.2 mm) opens at the top. n = 7 round support ribs are cylinder(a, b, r) between grid points, with drainage slots with filleted edges between them; underneath stand four feet (3 mm).





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	115.0 × 80.0 × 27.0 mm
Volume	37.0 cm³
Mass PLA (1.24 g/cm³)	45.8 g
Parts	1: `dish`
Colours / filaments	single colour (filament slot 1)
Kernel time (no cache)	1.56 s (38 nodes)

PARAMETERS (DEFAULT VALUES)

soap_dish() `l = 115` `b = 80` `h = 24` `wall = 2.2` `corner = 14` `n = 7` `foot_h = 3`

LANGUAGE FEATURES

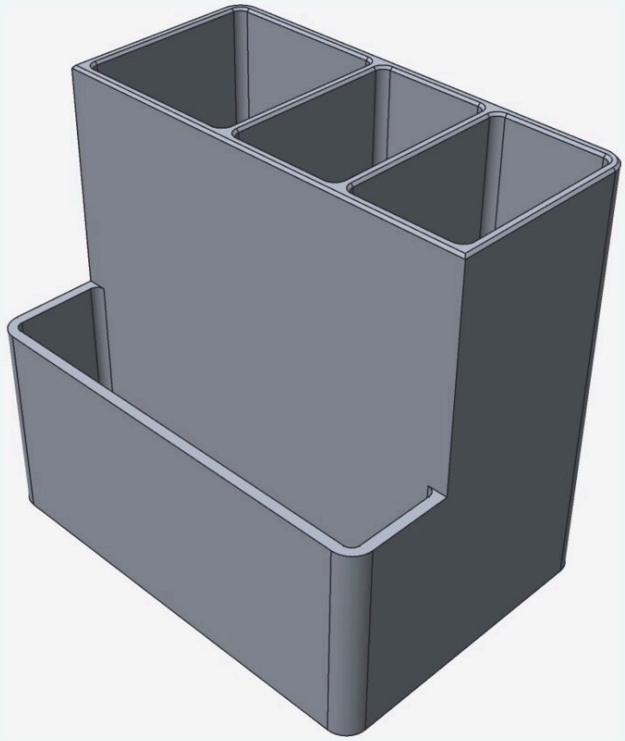
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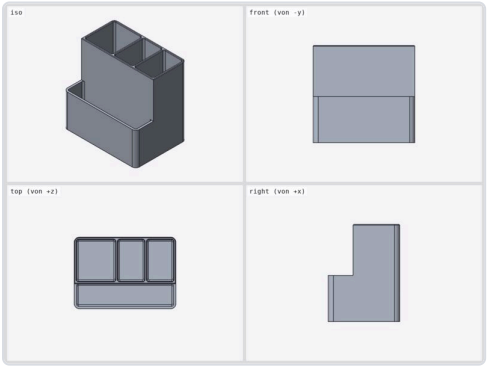
```
OPEN      cad tests/models/everyday/soap_dish.mcad
HEADLESS  cad render tests/models/everyday/soap_dish.mcad -o bild.png --views all
```

06 Pen box

Single part

Pen box with three tall compartments (widths 38 / 26 / 26 mm, depth 70 mm, height 95 mm) and a low front compartment (22 mm deep, 45 mm high), wall 2.4 mm. The compartment boundaries are indexed grid planes computed by a recursive function from the list of compartment widths. Four nested zone blocks fillet the vertical edges (outside r 5 mm, pockets r 2 mm) and the pocket floors and chamfer the rear top edges; the top edges of the front compartment stay sharp (kernel limit F6).





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	99.6 × 70.0 × 95.0 mm
Volume	115.2 cm³
Mass PLA (1.24 g/cm³)	142.8 g
Parts	1: box_
Colours / filaments	single colour (filament slot 1)
Kernel time (no cache)	1.20 s (41 nodes)

PARAMETERS (DEFAULT VALUES)

pen_box() `compartments = [38, 26, 26]` `depth = 70` `h = 95` `wall = 2.4` `floor_t = 3` `front_t = 22` `front_h = 45` `corner = 5` `corner_inner = 2`

LANGUAGE FEATURES

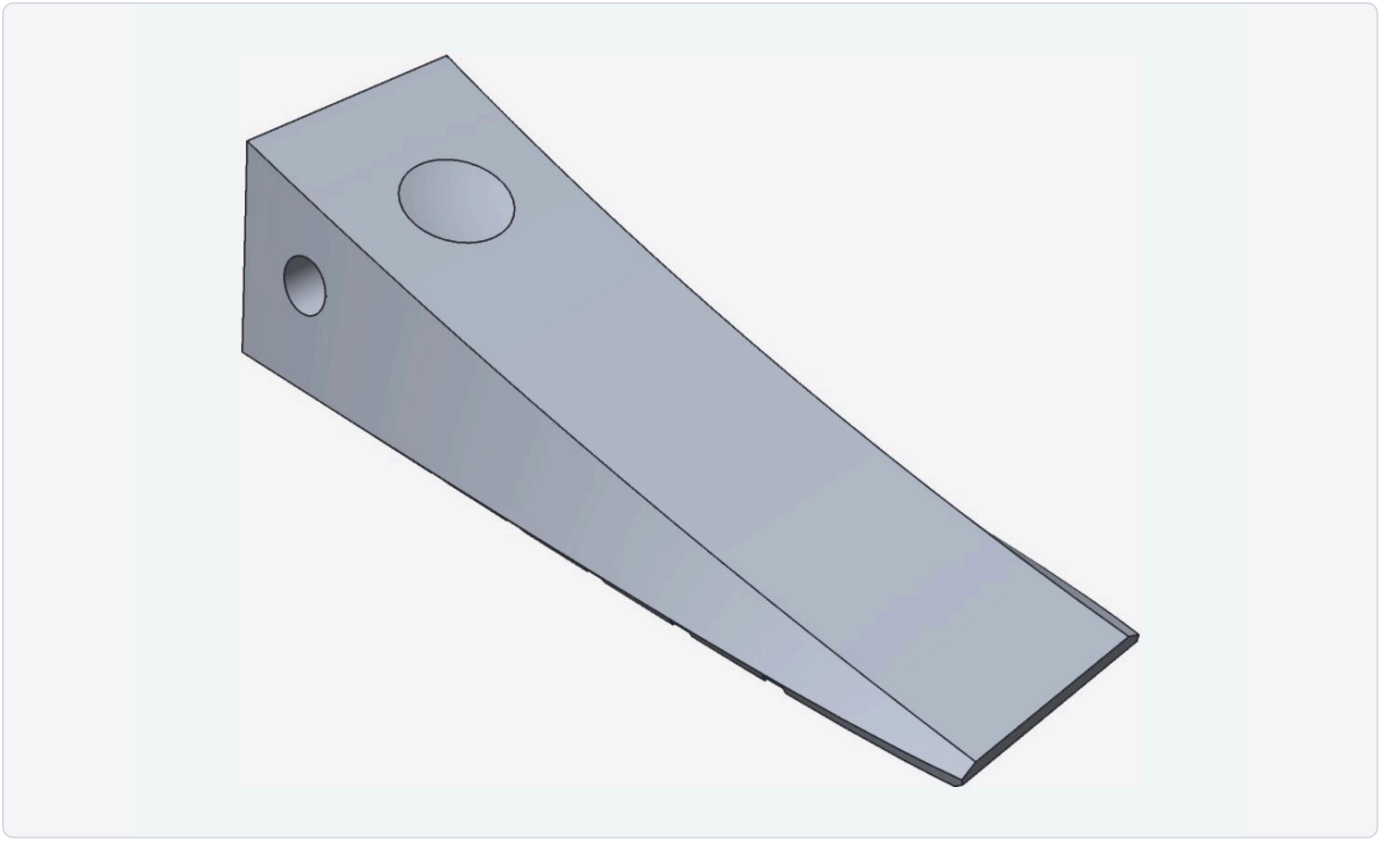
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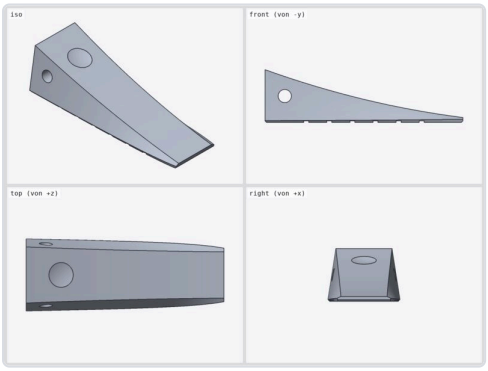
```
OPEN      cad tests/models/everyday/pen_box.mcad
HEADLESS  cad render tests/models/everyday/pen_box.mcad -o bild.png --views all
```

07 Door stop

Single part

Wedge 120 mm long, 44 mm wide and 32 mm high at the back. The body is a loft through three trapezoid profiles (height at the back, middle 14 mm, tip 3 mm), laid flat with rotate([90, 0, 90]); its bottom edge is chamfered beforehand via a zone on the bottom plane. On top there is a finger recess, underneath 6 transverse grooves against slipping, at the back a hanging hole Ø 8 mm.





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	120.0 × 43.6 × 32.0 mm
Volume	64.2 cm³
Mass PLA (1.24 g/cm³)	79.7 g
Parts	1: `stopper`
Colours / filaments	single colour (filament slot 1)
Kernel time (no cache)	0.29 s (22 nodes)

PARAMETERS (DEFAULT VALUES)

door_stop() `l = 120` `b = 44` `h_back = 32` `h_mid = 14` `tip = 3` `grooves = 6` `hole_d = 8`

LANGUAGE FEATURES

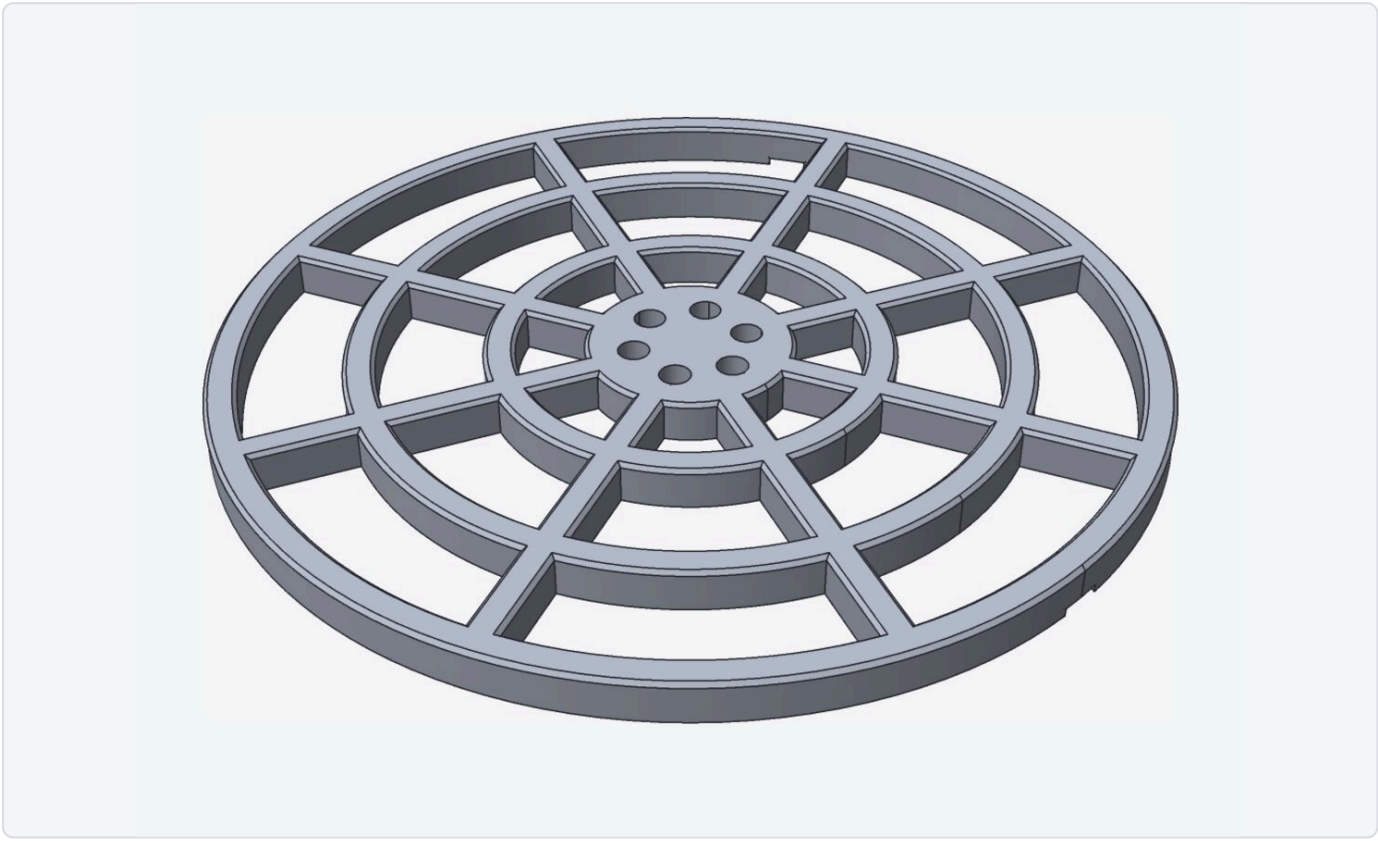
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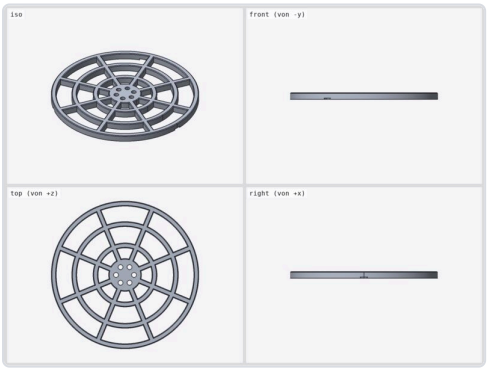
```
OPEN      cad tests/models/everyday/door_stop.mcad
HEADLESS  cad render tests/models/everyday/door_stop.mcad -o bild.png --views all
```

08 Pot trivet

Single part

Trivet Ø 180 mm, 8 mm high: three concentric rings (centre radii 36 / 62 / 87 mm, web 6 mm) as rotate_extrude of a rectangle in a for loop, connected by 8 spokes and a centre disc (r 20 mm). All top edges of rings and spokes are chamfered in one block via a zone on the top plane; underneath there are three recesses for rubber feet Ø 10 mm.





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	180.0 × 180.0 × 8.0 mm
Volume	83.3 cm³
Mass PLA (1.24 g/cm³)	103.2 g
Parts	1: u
Colours / filaments	single colour (filament slot 1)
Kernel time (no cache)	5.36 s (63 nodes)

PARAMETERS (DEFAULT VALUES)

trivet() `d = 180` `h = 8` `web = 6` `rings = [36, 62, 87]` `spokes = 8` `hub_r = 20` `foot_d = 10`

LANGUAGE FEATURES

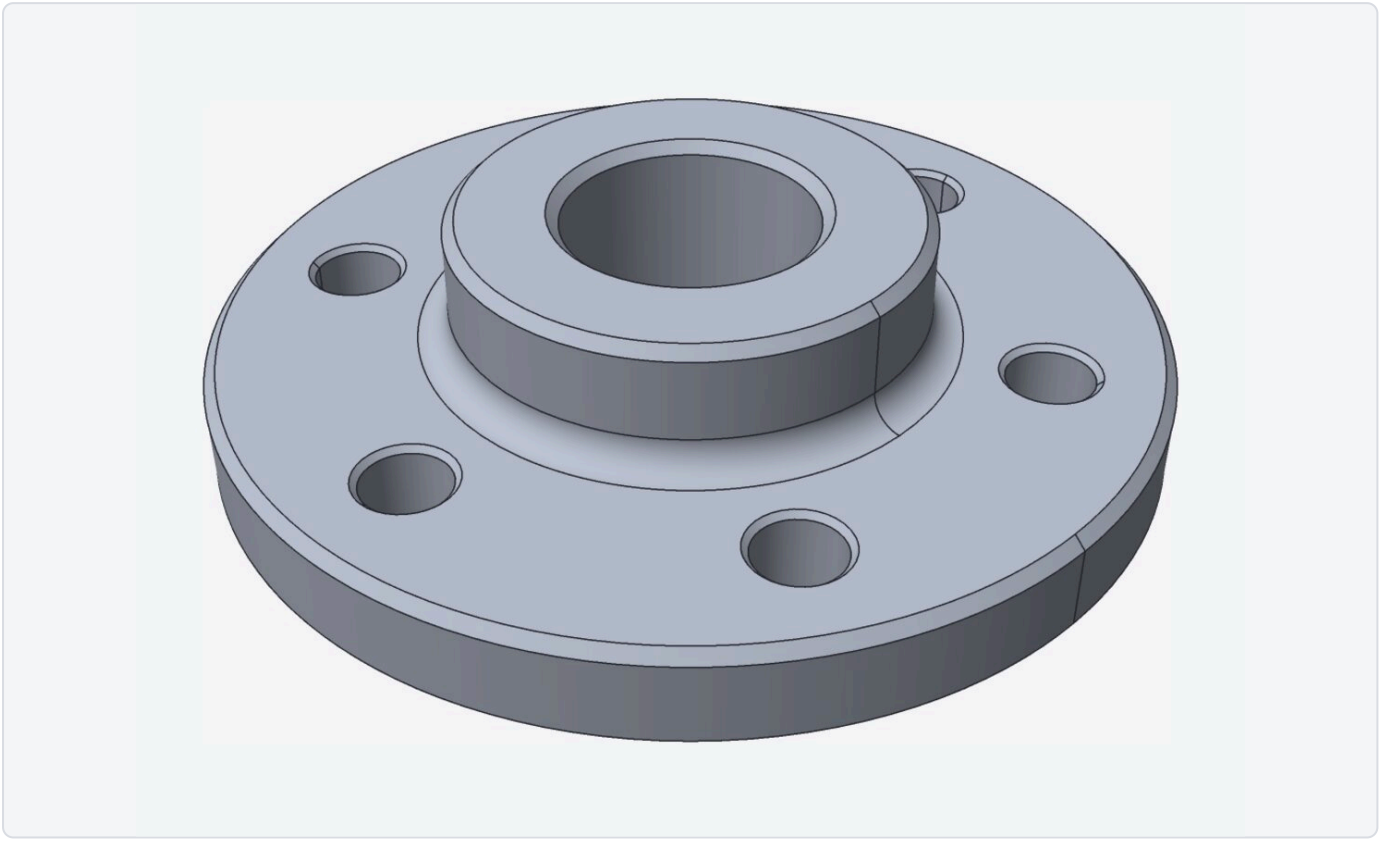
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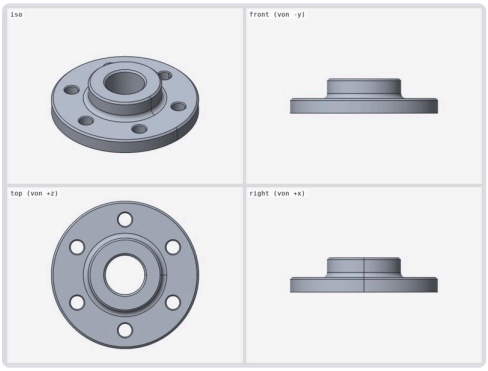
```
OPEN      cad tests/models/everyday/trivet.mcad
HEADLESS  cad render tests/models/everyday/trivet.mcad -o bild.png --views all
```

09 Flange plate

Single part

Flange Ø 120 mm, 12 mm thick, with collar Ø 60 mm (height 28 mm) and bore Ø 32 mm. The cross-section is a polygon from a 2D grid, revolved with rotate_extrude; on the bolt circle Ø 90 mm lie 6 through holes Ø 11 mm (M10), plus a sealing groove 1.5 mm deep. The transition collar/disc is filleted by a dedicated module used as a zone (ring_zone), bore and outer edges are chamfered via volume zones.





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	120.0 × 120.0 × 28.0 mm
Volume	150.3 cm³
Mass PLA (1.24 g/cm³)	186.4 g
Parts	1: f
Colours / filaments	single colour (filament slot 1)
Kernel time (no cache)	1.01 s (47 nodes)

PARAMETERS (DEFAULT VALUES)

flange() `d_outer = 120` `t = 12` `d_collar = 60` `h_collar = 28` `d_bore = 32` `bcd = 90` `n = 6` `d_hole = 11` `groove_t = 1.5`

LANGUAGE FEATURES

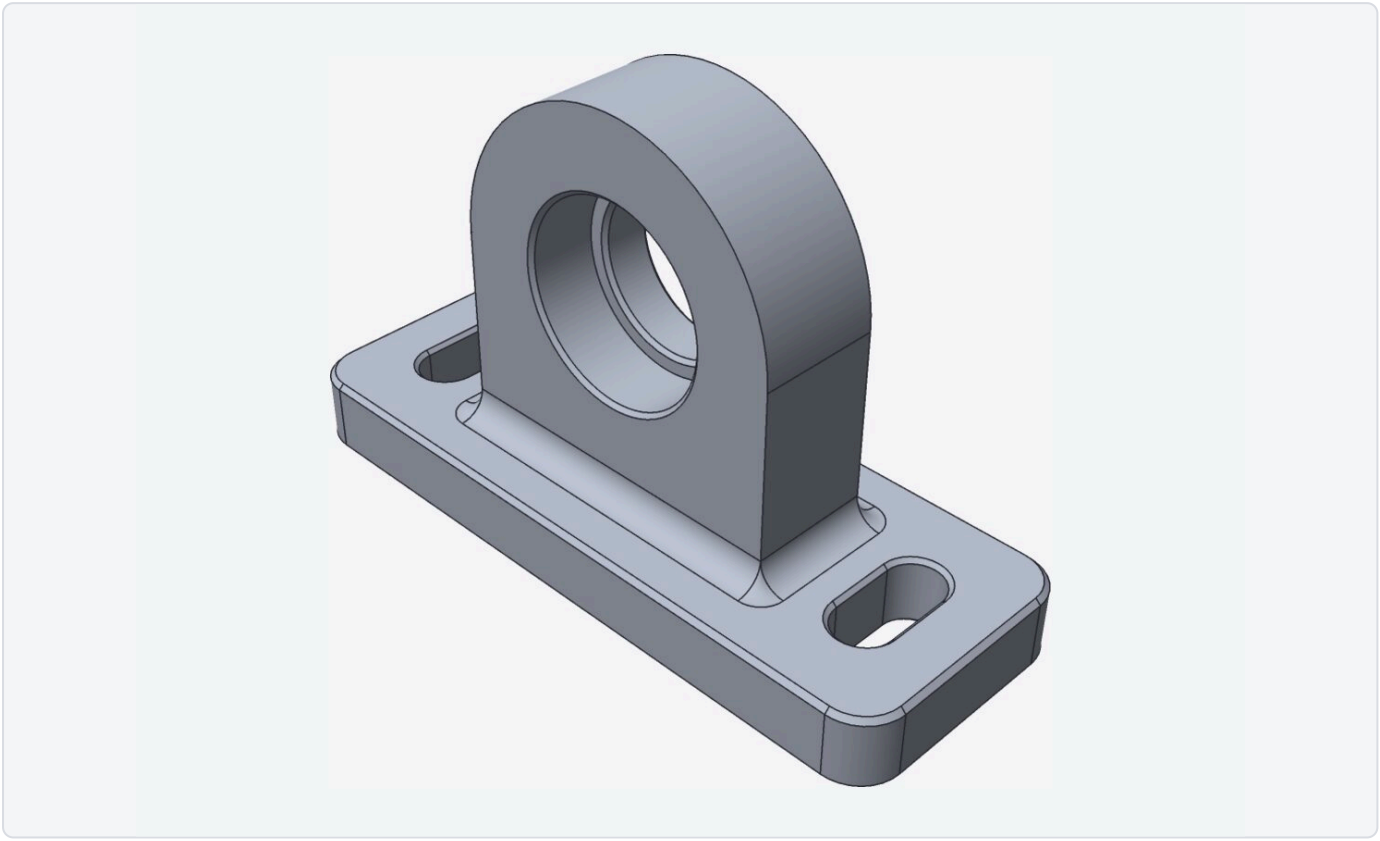
grid · anchor · anchor with dir · named instances · for · polygon · rotate_extrude · fillet zones · chamfer zones · difference(fillet/chamfer)

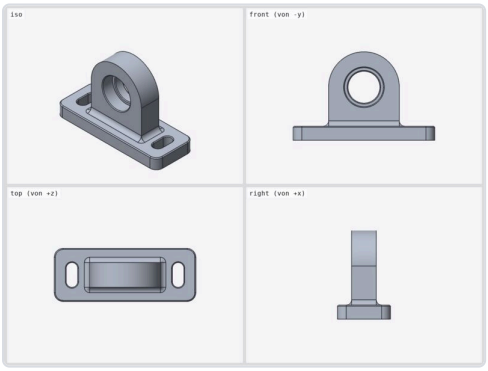
```
OPEN      cad tests/models/everyday/flange.mcad
HEADLESS  cad render tests/models/everyday/flange.mcad -o bild.png --views all
```

10 Pillow block 608

Single part

Bearing block for a 608 deep-groove ball bearing (22 × 8 mm), axis height 30 mm. The base plate 80 × 30 × 8 mm has two slots for M6 (6.5 × 14 mm, helper module) for alignment; the bearing eye (r 20 mm, 14 mm wide) is a box with a tangentially adjoining cylinder(a, b, r) and has a stop shoulder Ø 18 mm. The fillets around the eye are created via zones on grid lines in x, y and z.





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	80.0 × 30.0 × 50.0 mm
Volume	34.4 cm³
Mass PLA (1.24 g/cm³)	42.7 g
Parts	1: `block`
Colours / filaments	single colour (filament slot 1)
Kernel time (no cache)	0.82 s (38 nodes)

PARAMETERS (DEFAULT VALUES)

pillow_block() `l = 80` `b = 30` `t = 8` `axis_h = 30` `eye_r = 20` `eye_w = 14` `bearing_d = 22` `bearing_b = 8` `shoulder_d = 18` `slot_size = [6.5, 14]`

LANGUAGE FEATURES

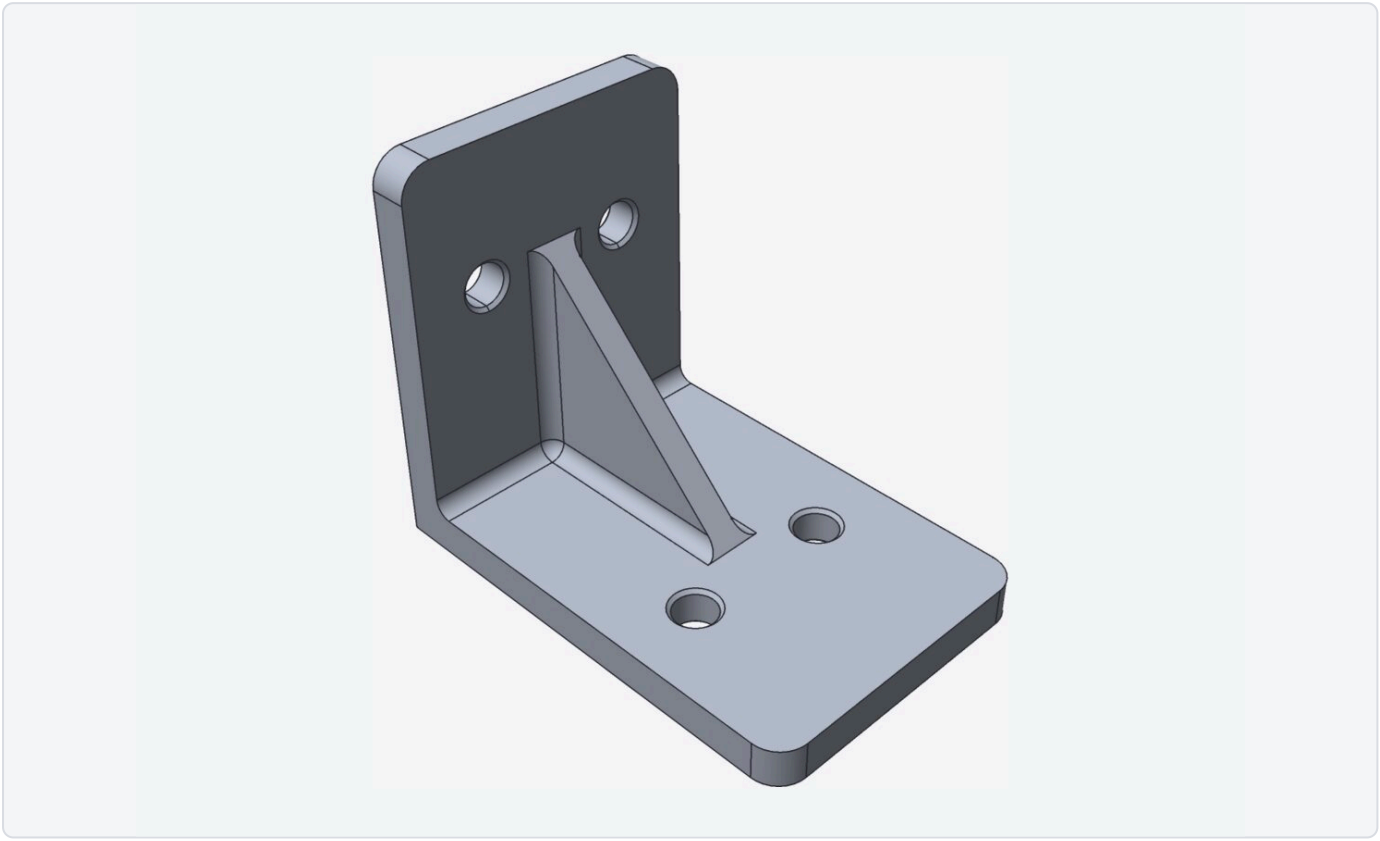
grid · anchor · anchor with dir · named instances · for · fillet zones · chamfer zones · zone · difference(fillet/chamfer) · box(a, b) · cylinder(a, b, r)

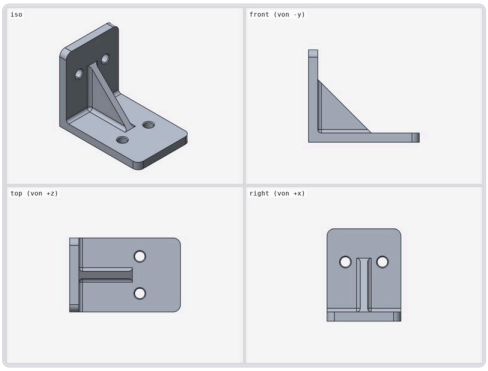
```
OPEN      cad tests/models/everyday/pillow_block.mcad
HEADLESS  cad render tests/models/everyday/pillow_block.mcad -o bild.png --views all
```

11 Mounting bracket with rib

Single part

Bracket with legs of 60 and 50 mm, 40 mm wide and 5 mm thick, with n_ribs = 1 stiffening rib (4 mm thick, 30 mm long). The ribs are extruded triangles on indexed grid planes; inner edge and rib joints are filleted in one zone block, so that the corner transitions run cleanly. The holes Ø 5.5 mm (M5) are chamfered with difference(chamfer = ...).





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	60.0 × 40.0 × 50.0 mm
Volume	22.2 cm³
Mass PLA (1.24 g/cm³)	27.6 g
Parts	1: `w`
Colours / filaments	single colour (filament slot 1)
Kernel time (no cache)	0.58 s (31 nodes)

PARAMETERS (DEFAULT VALUES)

rib_bracket() `a = 60` `h = 50` `b = 40` `t = 5` `rib_t = 4` `rib_l = 30` `n_ribs = 1` `hole_d = 5.5`

LANGUAGE FEATURES

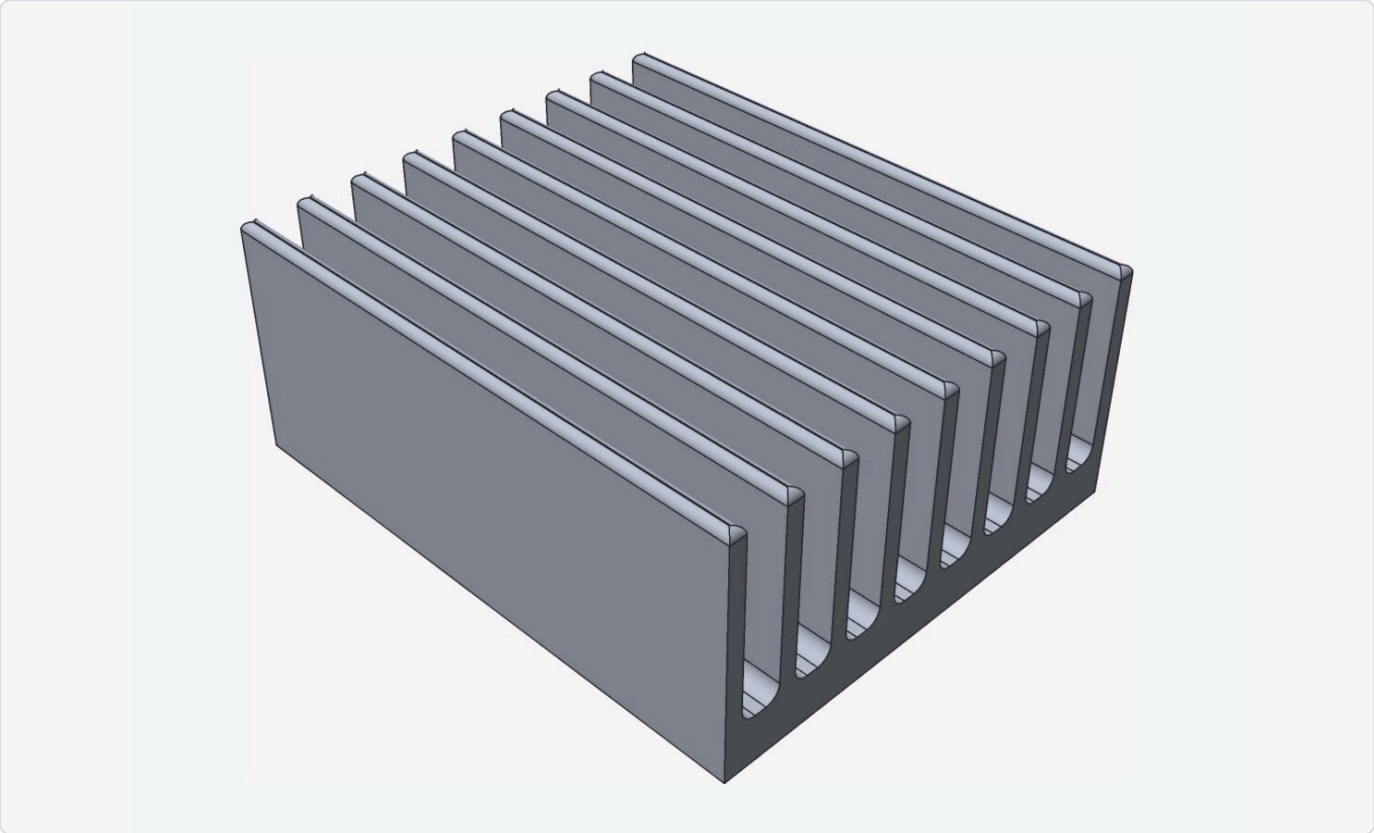
grid · anchor · anchor with dir · named instances · list comprehension · for · polygon · linear_extrude · fillet zones · zone · difference(fillet/chamfer) · box(a, b) · cylinder(a, b, r)

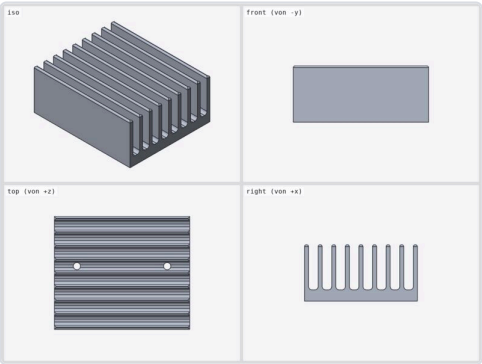
```
OPEN      cad tests/models/everyday/rib_bracket.mcad
HEADLESS  cad render tests/models/everyday/rib_bracket.mcad -o bild.png --views all
```

12 Heat sink

Single part

Extruded heat sink 60 × 50 mm, base plate 5 mm, total height 25 mm, with n = 9 fins of 1.8 mm each. The fin positions are indexed grid planes from a list; the fin roots are filleted via zones on grid lines in a for loop (r 1.5 mm), the tips almost fully rounded via a zone on the tip plane. Two M3 holes 40 mm apart lie between the fins, an anchor marks the heat source.





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	60.0 × 50.0 × 25.0 mm
Volume	34.7 cm³
Mass PLA (1.24 g/cm³)	43.0 g
Parts	1: hs
Colours / filaments	single colour (filament slot 1)
Kernel time (no cache)	1.05 s (48 nodes)

PARAMETERS (DEFAULT VALUES)

heat_sink() `l = 60` `b = 50` `base = 5` `h = 25` `n = 9` `fin_t = 1.8` `foot_r = 1.5` `hole_spacing = 40`

LANGUAGE FEATURES

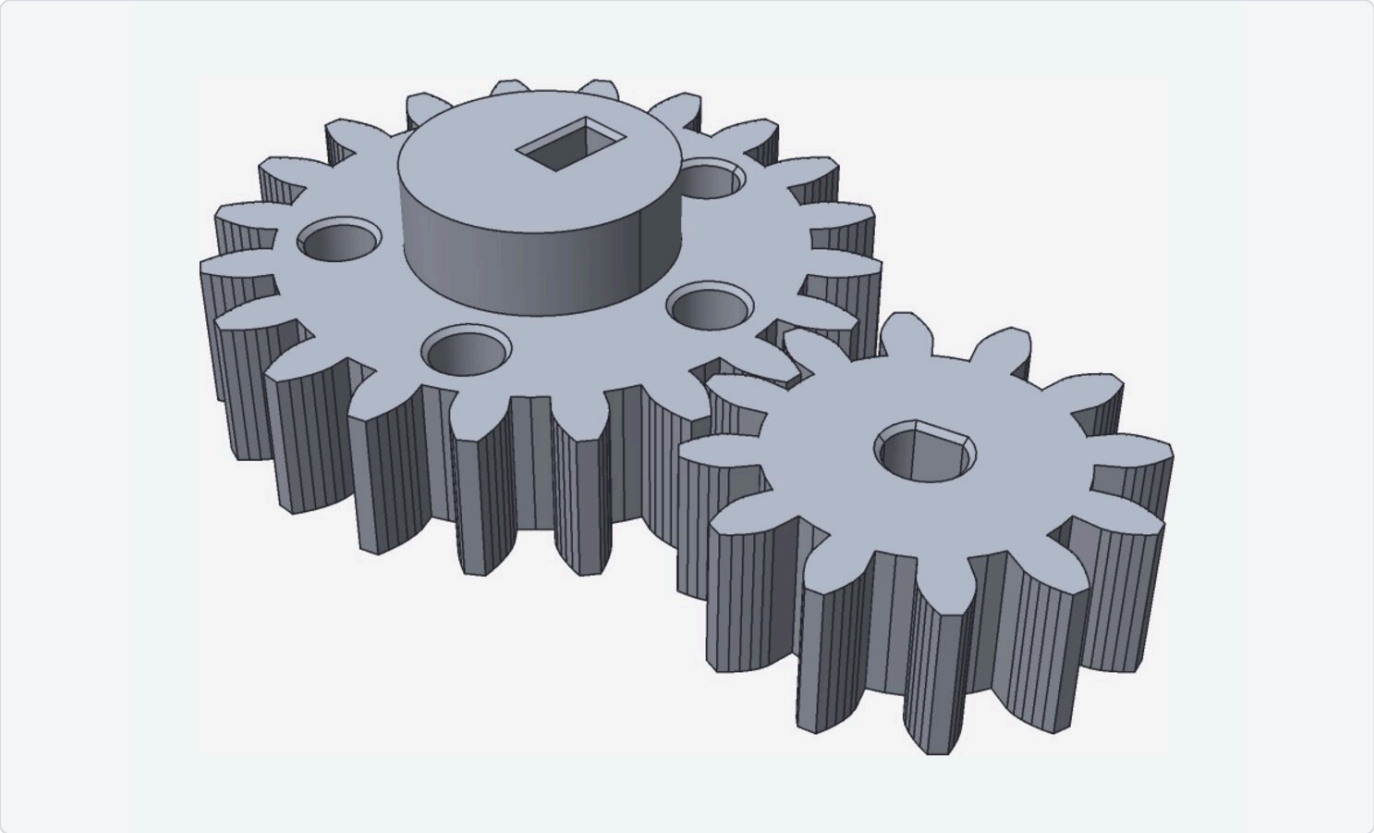
grid · indexed grid planes · anchor · anchor with dir · named instances · list comprehension · for · fillet zones · zone · box(a, b)

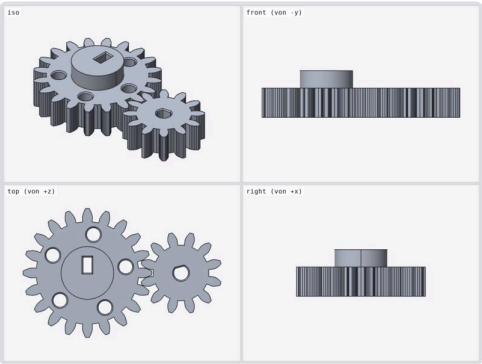
```
OPEN      cad tests/models/everyday/heat_sink.mcad
HEADLESS  cad render tests/models/everyday/heat_sink.mcad -o bild.png --views all
```

13 Spur gear pair

Assembly

Two meshing spur gears with module m = 2 mm and 20 and 12 teeth, centre distance 32 mm, width 10 mm. The involute flanks are created as a polygon from list functions (let, each, concat) and extruded linearly; the tooth thickness is slightly reduced for backlash. Gear 1 has a hub, keyway and lightening holes, gear 2 a D bore for a 5 mm motor shaft.





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	67.6 × 44.0 × 16.0 mm
Volume	16.6 cm³
Mass PLA (1.24 g/cm³)	20.6 g
Parts	2: gear1, gear2
Colours / filaments	single colour (filament slot 1)
Kernel time (no cache)	4.97 s (48 nodes)

PARAMETERS (DEFAULT VALUES)

File `m = 2` `z1 = 20` `z2 = 12` gear() `m = 2` `z = 20` `b = 10` `bore = 8` `hub_d = 0` `hub_h = 0` `key_b = 3` `key_t = 1.4` `d_flat = 0` `holes = 0`

LANGUAGE FEATURES

grid · anchor · anchor with dir · named instances · function · let · list comprehension · each · for · if / ?: · polygon · linear_extrude · difference(fillet/chamfer) · intersection

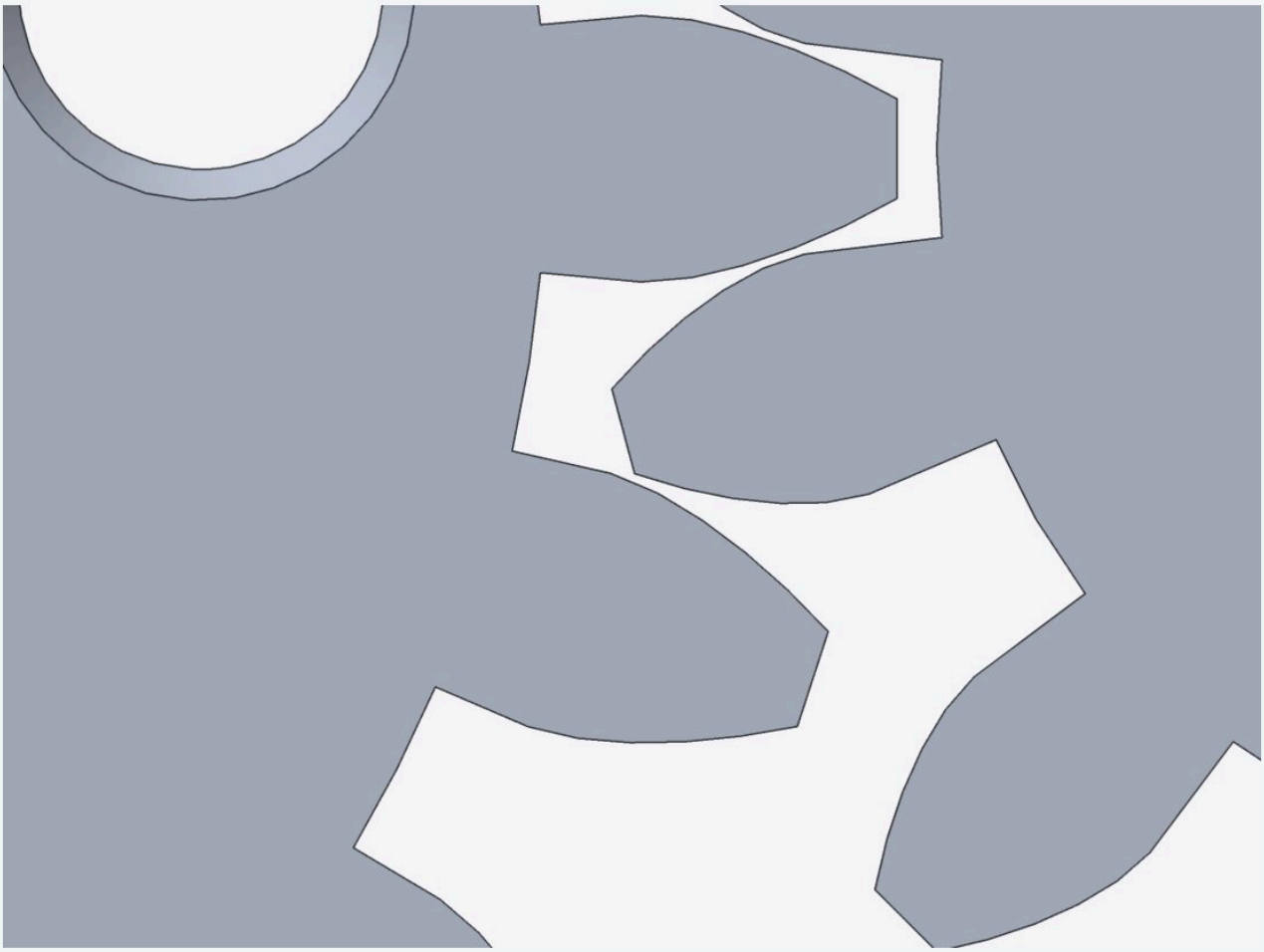
```
OPEN      cad tests/models/everyday/gear.mcad
HEADLESS  cad render tests/models/everyday/gear.mcad -o bild.png --views all
```

13 Spur gear pair – fit

Assembly

Part	Module	Bounding box (W × D × H) mm	Volume cm³	Mass PLA g
gear1	gear	44.0 × 44.0 × 16.0	12.55	15.6
gear2	gear	34.3 × 34.3 × 10.0	4.08	5.1

Pair (`measure --clearance`)	Distance	Meaning
gear1 ↔ gear2	0.116 mm	Backlash
`measure --interference`	no collision (tolerance 0.001 mm³)	



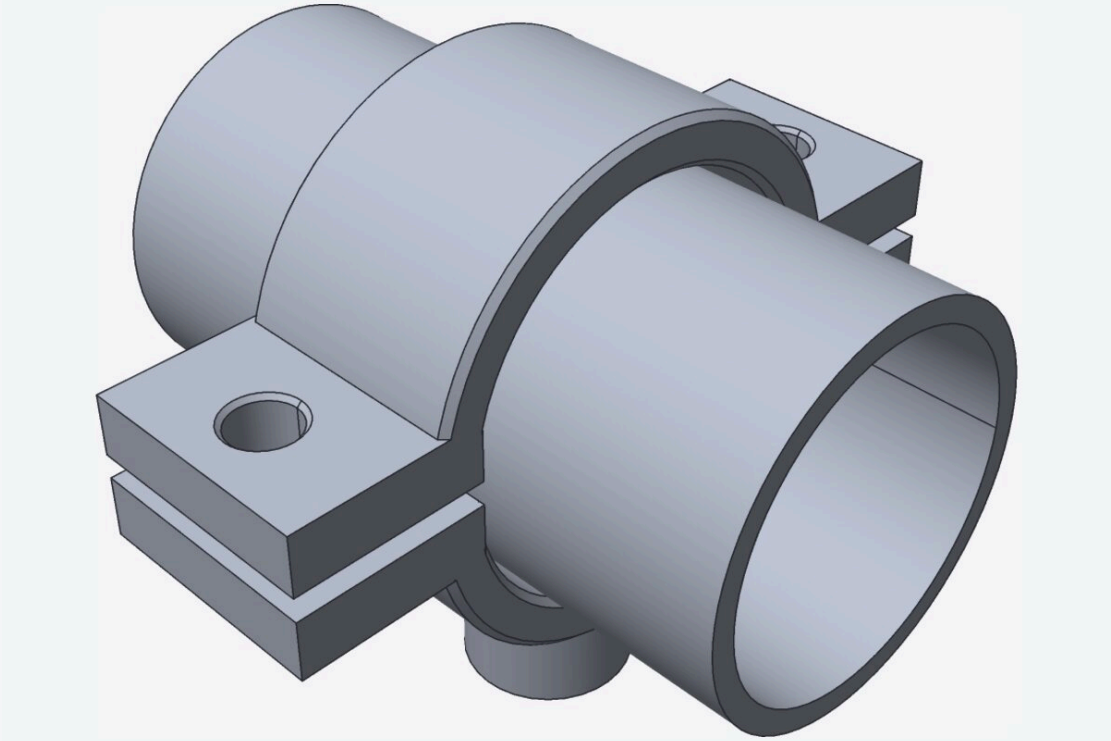
Top view, magnified 6× on the narrowest point of the mesh: the gap between the flanks is the measured backlash.

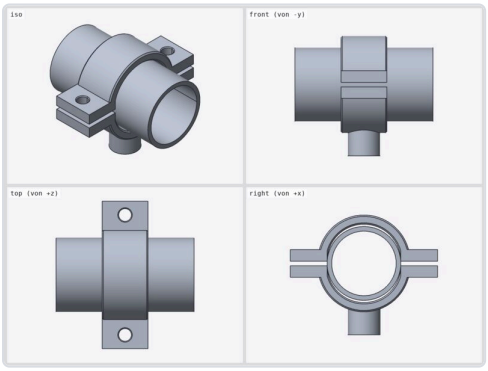
```
CHECK  cad measure tests/models/everyday/gear.mcad --interference --clearance gear1 gear2
```

14 Pipe clamp

Assembly

Two-part clamp for a pipe Ø 32 mm: the half rings (4 mm thick, 20 mm wide) are rotate_extrude(angle = 180) of a rectangle, with lugs and holes Ø 5.5 mm (M5); the lower part (mirror) carries a boss with a pilot hole for M8. Upper part, lower part and a test pipe are parts in a jointly rotated block. In the open state there is a 2 mm gap at the lugs.





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	60.0 × 64.0 × 52.0 mm
Volume	25.9 cm³
Mass PLA (1.24 g/cm³)	32.1 g
Parts	3: `top`, `bottom`, `pipe`
Colours / filaments	single colour (filament slot 1)
Kernel time (no cache)	1.79 s (51 nodes)

PARAMETERS (DEFAULT VALUES)

File `pipe_d = 32` `b = 20` half_clamp() `pipe_d = 32` `t = 4` `b = 20` `tab = 12` `tab_t = 5` `gap = 2` `hole_d = 5.5` `stud = false`

LANGUAGE FEATURES

grid · anchor · anchor with dir · named instances · for · if / ?: · rotate_extrude · chamfer zones · zone · difference(fillet/chamfer) · box(a, b) · cylinder(a, b, r) · mirror

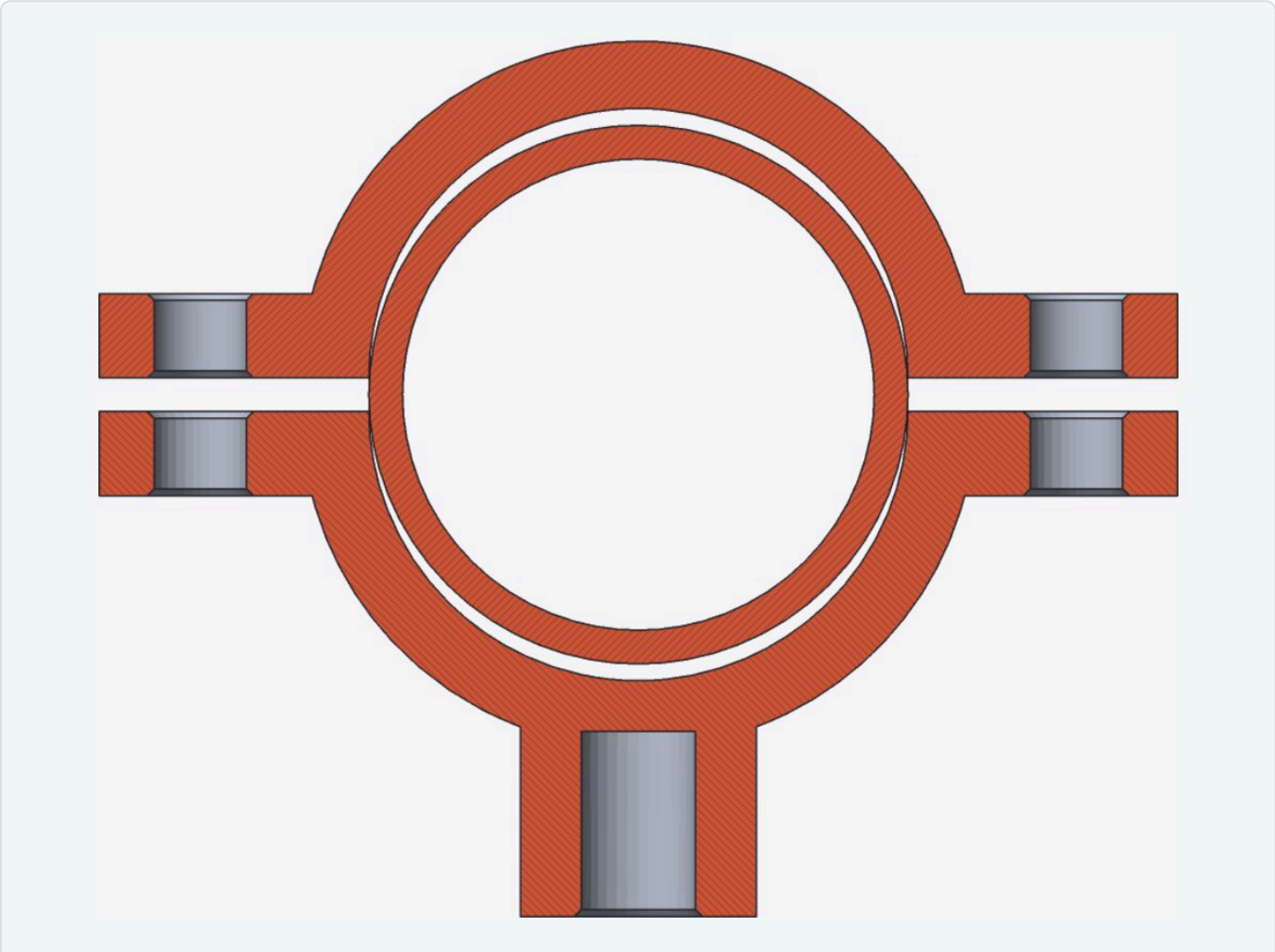
```
OPEN      cad tests/models/everyday/pipe_clamp.mcad
HEADLESS  cad render tests/models/everyday/pipe_clamp.mcad -o bild.png --views all
```

14 Pipe clamp – fit

Assembly

Part	Module	Bounding box (W × D × H) mm	Volume cm³	Mass PLA g
top	half_clamp	20.0 × 64.0 × 20.0	6.69	8.3
bottom	half_clamp	20.0 × 64.0 × 30.0	7.87	9.8
pipe	test_pipe	60.0 × 32.0 × 32.0	11.31	14.0

Pair (measure --clearance)	Distance	Meaning
top ↔ bottom	2.000 mm	Clamping gap at the lugs
top ↔ pipe	0.035 mm	Pipe touches the upper part
bottom ↔ pipe	0.035 mm	Pipe touches the lower part
measure --interference	no collision (tolerance 0.001 mm³)	



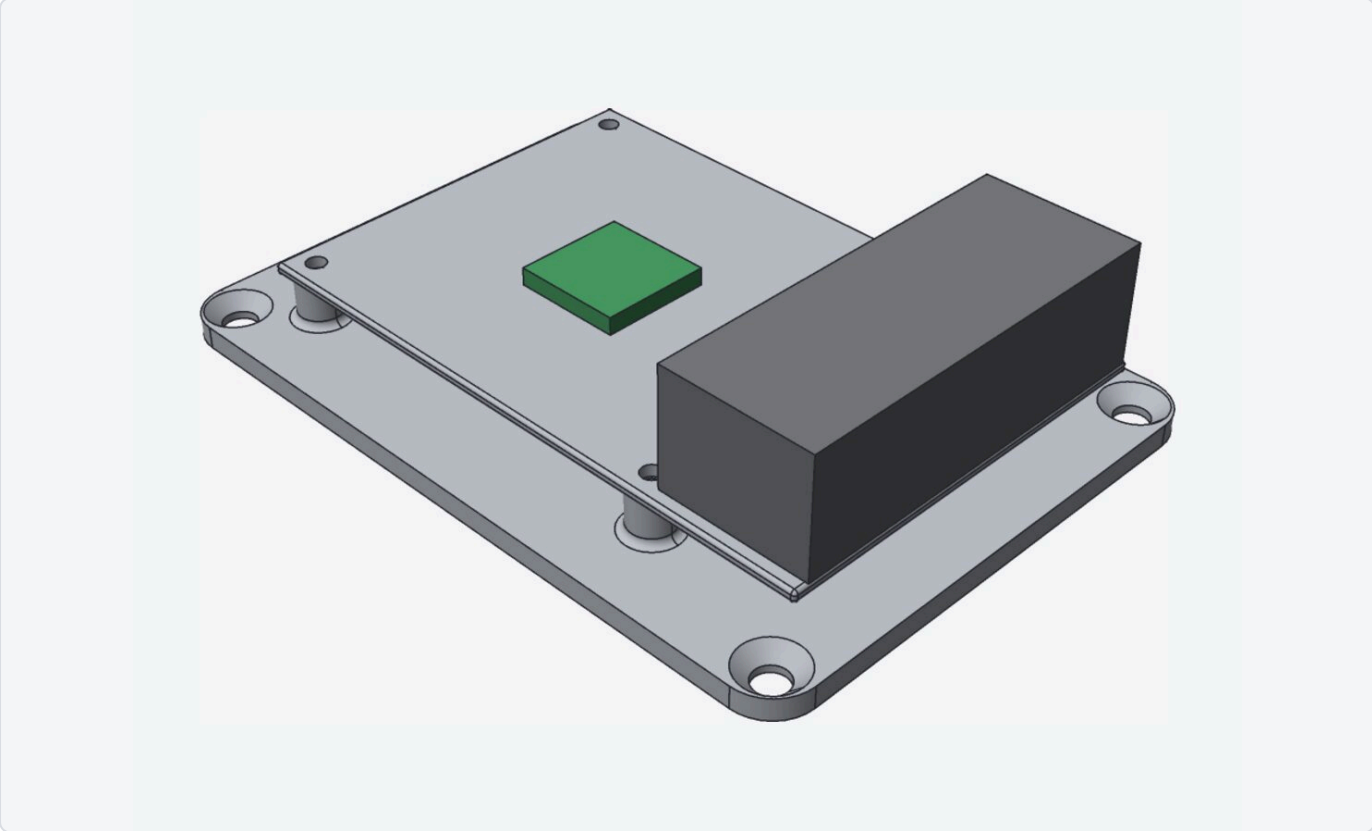
Section x = 10 (middle of the clamp, across the pipe axis), view from +x: half rings with lugs and M5 holes, clamping gap 2 mm, boss with pilot hole for M8, test pipe.

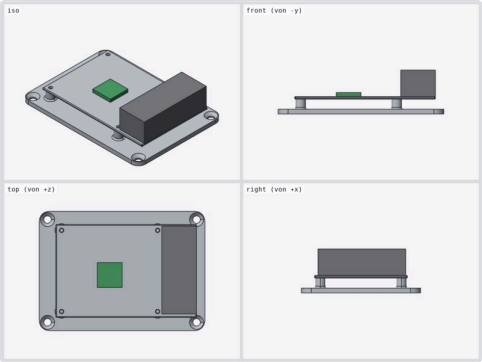
```
CHECK  cad measure tests/models/everyday/pipe_clamp.mcad --interference --clearance top bottom
```

15 Raspberry Pi mounting plate

Assembly

Plate 100 × 72 × 3 mm for a Raspberry Pi (hole pattern 58 × 49 mm, M2.5). The standoffs (6 mm) are a separate module with an anchor, placed in the loop as named instances boss[i] and attached with union(fillet = …). In addition there are 6 ventilation slots, four M4 mounting holes with countersink and filleted corners (r 6 mm). A coloured board dummy 85 × 56 × 1.6 mm sits as a second part on the anchor of the plate.





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	100.0 × 72.0 × 26.6 mm
Volume	46.0 cm³
Mass PLA (1.24 g/cm³)	57.1 g
Parts	2: `plate`, `pi`
Colours / filaments	Slot 1 default 34.3 g Slot 2 green 0.7 g Slot 3 gray 22.1 g
Kernel time (no cache)	1.85 s (118 nodes)

PARAMETERS (DEFAULT VALUES)

pi_plate() `l = 100` `b = 72` `t = 3` `corner = 6` `height = 6` `margin_x = 10` `margin_y = 8` `slots = 6`

LANGUAGE FEATURES

grid · anchor · anchor with dir · named instances · list comprehension · for · if / ?: · fillet zones · zone · union(fillet) · $fillet · box(a, b) · cone (r1, r2) · color

```
OPEN      cad tests/models/everyday/pi_plate.mcad
HEADLESS  cad render tests/models/everyday/pi_plate.mcad -o bild.png --views all
```

15 Raspberry Pi mounting plate – fit

Assembly

Part	Module	Bounding box (W × D × H) mm	Volume cm³	Mass PLA g
plate	pi_plate	100.0 × 72.0 × 9.0	20.15	25.0
pi	pi_board	85.0 × 56.0 × 17.6	25.90	32.1

Pair (measure --clearance)	Distance	Meaning
plate ↔ pi	0.000 mm	Board rests on the bosses
measure --interference	no collision (tolerance 0.001 mm³)	

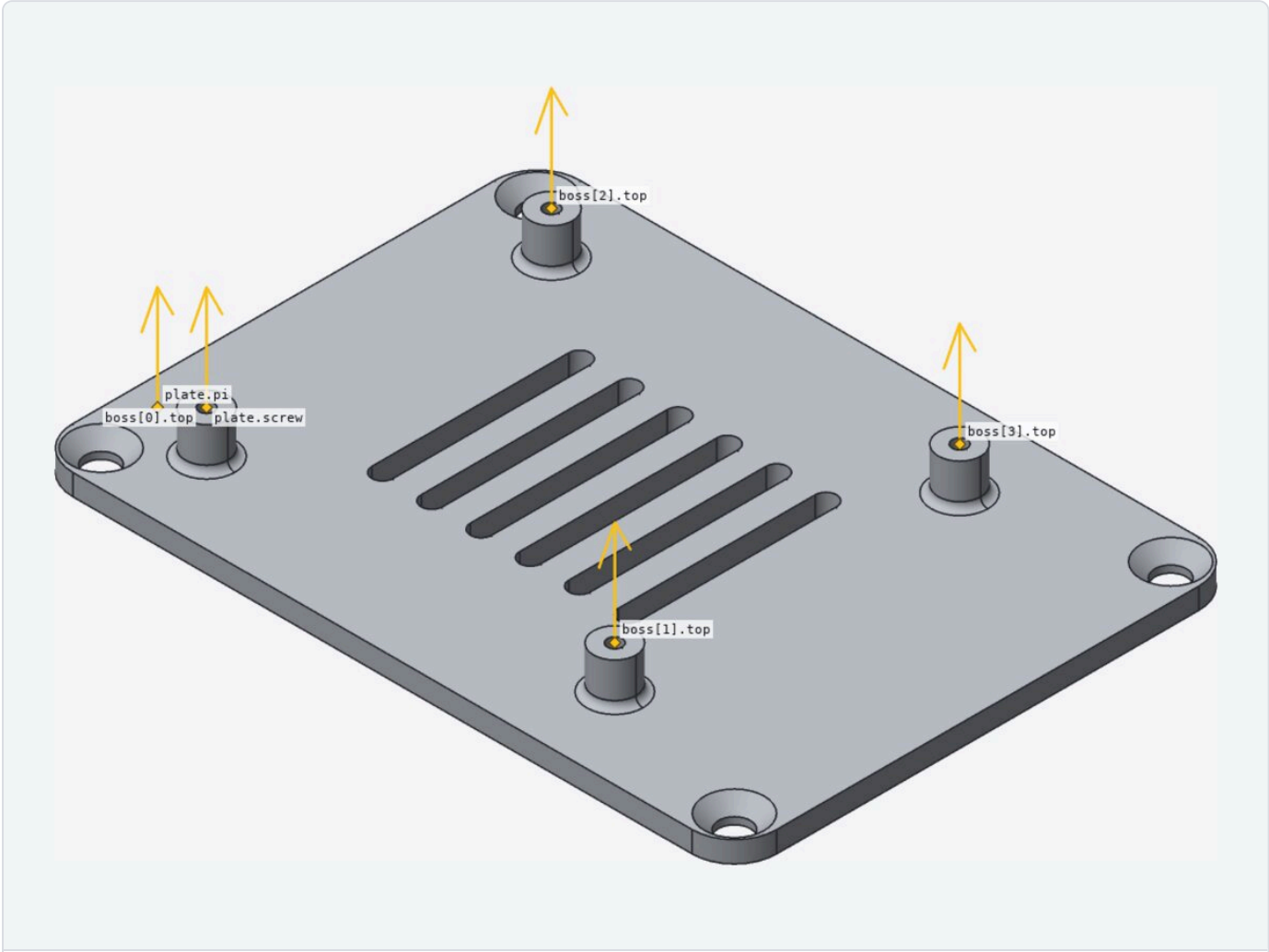


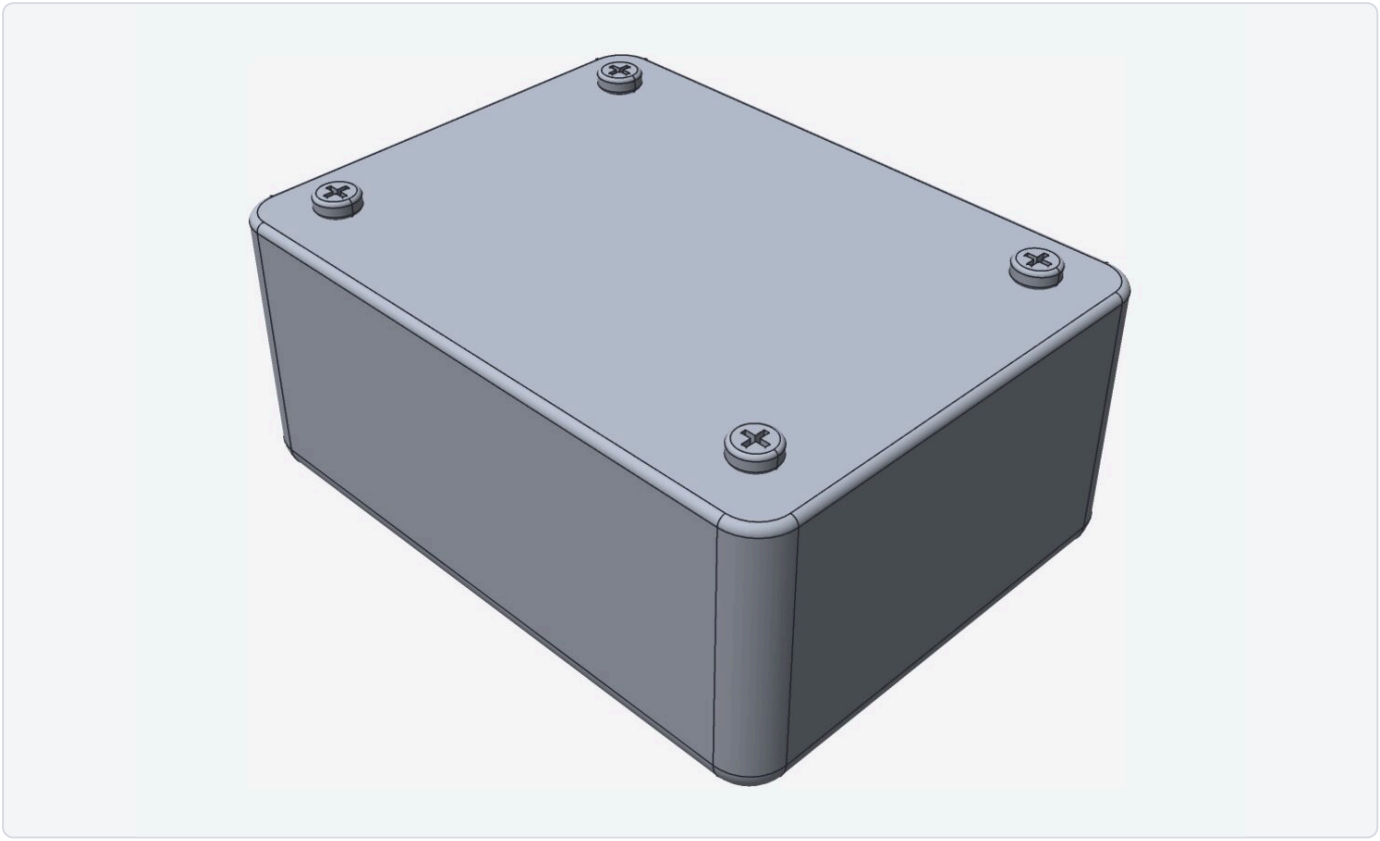
Plate with anchors shown (yellow): the anchors top of the bosses carry the board.

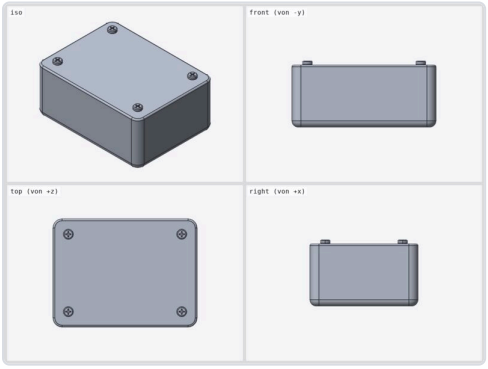
```
CHECK  cad measure tests/models/everyday/pi_plate.mcad --interference --clearance plate pi
```

16 Electronics enclosure

Assembly

Enclosure for a 70 × 50 mm board: inside 76 × 56 mm, height 32 mm, wall 2 mm, corners r 5 mm. The base is a filleted box opened by shell; four screw bosses (pilot hole Ø 2.5 mm) are attached as boss[i] with union(fillet = …). The lid (2.5 mm, centring rim 3 mm, clearance 0.2 mm) sits on the anchor lid_seat, the M3 screws are parts s[0] … s[3]. Dimensions are in dimensions.mcad (include), parts in parts.mcad (use); explosion > 0 pulls the assembly apart.





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	80.0 × 60.0 × 36.7 mm
Volume	42.1 cm³
Mass PLA (1.24 g/cm³)	52.2 g
Parts	6: `base`, `lid`, `s[0]`, `s[1]`, `s[2]`, `s[3]`
Colours / filaments	single colour (filament slot 1)
Kernel time (no cache)	3.67 s (126 nodes)

PARAMETERS (DEFAULT VALUES)

File `inner_l = 76` `inner_w = 56` `height = 32` `wall = 2` `corner = 5` `lid_t = 2.5` `rim_h = 3` `clearance = 0.2` `boss_d = 7` `core_d = 2.5` `through_d = 3.4` `screw_l = 12` `boss_gap = 3` `explosion = 0`

LANGUAGE FEATURES

grid · anchor · anchor with dir · named instances · include · use · function · let · for · if / ?: · shell · fillet zones · zone · difference(fillet/chamfer) · union(fillet) · $fillet · box(a, b) · cylinder(a, b, r) · cone (r1, r2)

```
OPEN      cad tests/models/everyday/housing/housing.mcad
HEADLESS  cad render tests/models/everyday/housing/housing.mcad -o bild.png --views all
```

16 Electronics enclosure – fit

Assembly

Part	Module	Bounding box (W × D × H) mm	Volume cm³	Mass PLA g
base	base	80.0 × 60.0 × 32.0	29.18	36.2
lid	lid	80.0 × 60.0 × 5.5	12.50	15.5
s[0]	screw	5.5 × 5.5 × 14.2	0.10	0.1
s[1]	screw	5.5 × 5.5 × 14.2	0.10	0.1
s[2]	screw	5.5 × 5.5 × 14.2	0.10	0.1
s[3]	screw	5.5 × 5.5 × 14.2	0.10	0.1

Pair (measure --clearance)	Distance	Meaning
base ↔ lid	0.000 mm	Lid rests on the base
s[0] ↔ lid	0.000 mm	Screw head rests on the lid
s[0] ↔ base	0.050 mm	Shank in the pilot hole
measure --interference	no collision (tolerance 0.001 mm³)	



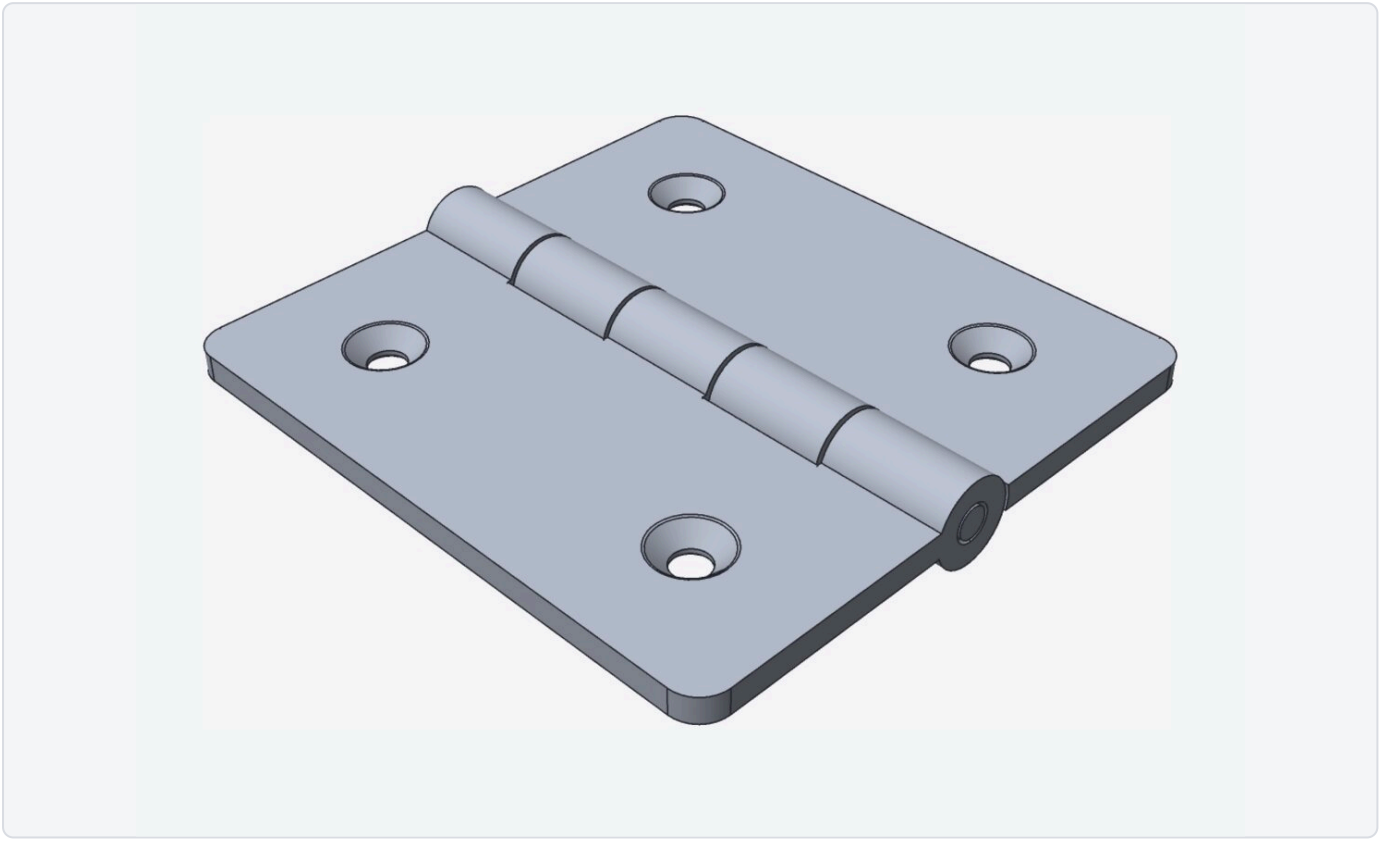
Section y = 8.5 (through the front bosses), view from the front: screw in the boss, lid with centring rim on the base.

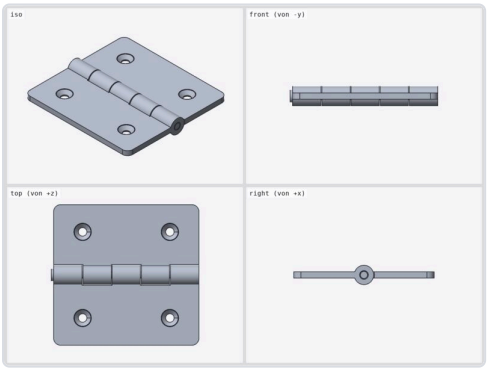
```
CHECK  cad measure tests/models/everyday/housing/housing.mcad --interference --clearance base lid
```

17 Hinge

Assembly

Printable 60 mm hinge made of leaf A, leaf B and pin with 5 knuckles (r 4 mm, leaf thickness 2.5 mm). The knuckles of both leaves come from the same function, so that the clearance of 0.4 mm is the same everywhere; the gaps are created by for with a step. Leaf A has the 3 mm hole (pin glued in), leaf B 3.4 mm (rotates freely), the pin Ø 2.95 mm. The variable angle = 180° rotates leaf B about the axis.





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	61.0 × 58.0 × 8.0 mm
Volume	10.2 cm³
Mass PLA (1.24 g/cm³)	12.6 g
Parts	3: a, b, pin
Colours / filaments	single colour (filament slot 1)
Kernel time (no cache)	2.27 s (60 nodes)

PARAMETERS (DEFAULT VALUES)

File `length = 60` `knuckles = 5` `knuckle_r = 4` `leaf_t = 2.5` `clearance = 0.4` `bore_a = 3` `bore_b = 3.4` `pin_d = 2.95` `angle = 180`

LANGUAGE FEATURES

grid · indexed grid planes · anchor · anchor with dir · named instances · include · use · function · for · if / ?: · fillet zones · zone · difference(fillet/chamfer) · box(a, b) · cylinder(a, b, r) · cone (r1, r2)

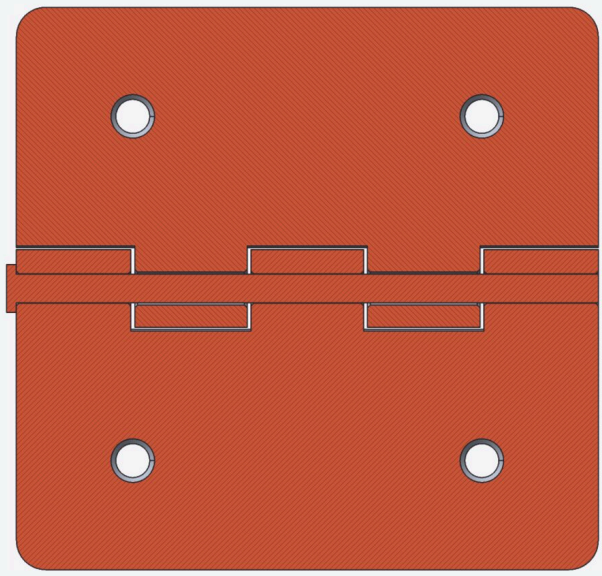
```
OPEN      cad tests/models/everyday/hinge/hinge.mcad
HEADLESS  cad render tests/models/everyday/hinge/hinge.mcad -o bild.png --views all
```

17 Hinge – fit

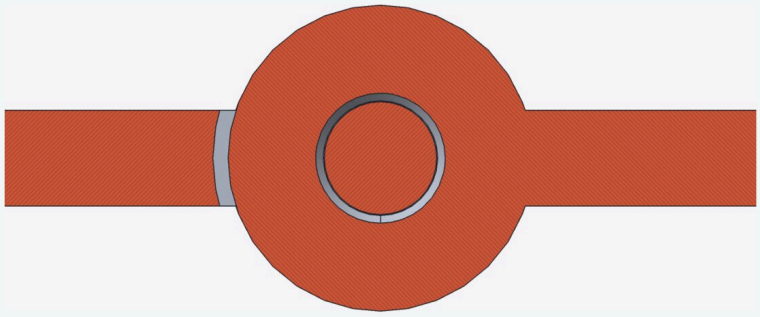
Assembly

Part	Module	Bounding box (W × D × H) mm	Volume cm³	Mass PLA g
a	leaf	60.0 × 33.0 × 8.0	5.15	6.4
b	leaf	60.0 × 33.0 × 8.0	4.57	5.7
pin	pin	61.0 × 5.0 × 5.0	0.43	0.5

Pair (`measure --clearance`)	Distance	Meaning
a ↔ b	0.397 mm	Clearance of the knuckles
pin ↔ b	0.223 mm	Bearing clearance
pin ↔ a	0.000 mm	Pin head touches
`measure --interference`	no collision (tolerance 0.001 mm³)	



Section z = 0 (mid-plane of the leaves), top view: knuckles of A (bottom) and B (top) alternating with axial clearance, pin in the holes.



Section x = 18 (second knuckle, leaf B), view from +x, 3.5×: pin Ø 2.95 in the 3.4 hole of B; on the left leaf A with clearance.

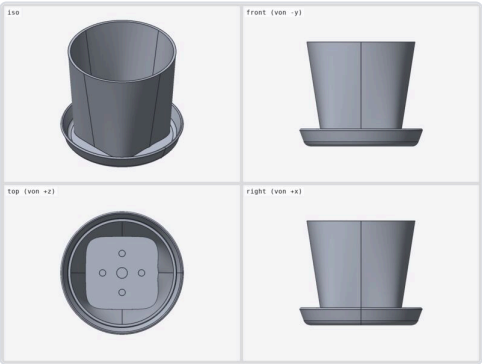
```
CHECK  cad measure tests/models/everyday/hinge/hinge.mcad --interference --clearance a b
```

18 Plant pot with saucer

Assembly

The pot goes by loft from a rounded square (72 mm, superellipse as a spline from a function) to the round opening Ø 100 mm, height 90 mm, wall 2.2 mm, bottom 3 mm; it stands on 4 mm high feet and has drainage holes. The wall is the difference of two lofts. The saucer is a rotate_extrude, opened at the top with shell and filleted via zones. The pot stands with the anchor footing on the anchor floor of the saucer.





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	113.7 × 113.7 × 96.4 mm
Volume	107.2 cm³
Mass PLA (1.24 g/cm³)	132.9 g
Parts	2: saucer, pot
Colours / filaments	single colour (filament slot 1)
Kernel time (no cache)	1.15 s (56 nodes)

PARAMETERS (DEFAULT VALUES)

File `pot_bottom = 72` `pot_top_d = 100` `pot_h = 90` `wall = 2.2` `floor_t = 3` `foot_h = 4` `gap = 4`

LANGUAGE FEATURES

grid · anchor · anchor with dir · named instances · include · use · function · let · list comprehension · for · if / ?: · polygon · spline · rotate_extrude · loft · shell · fillet zones · $fillet

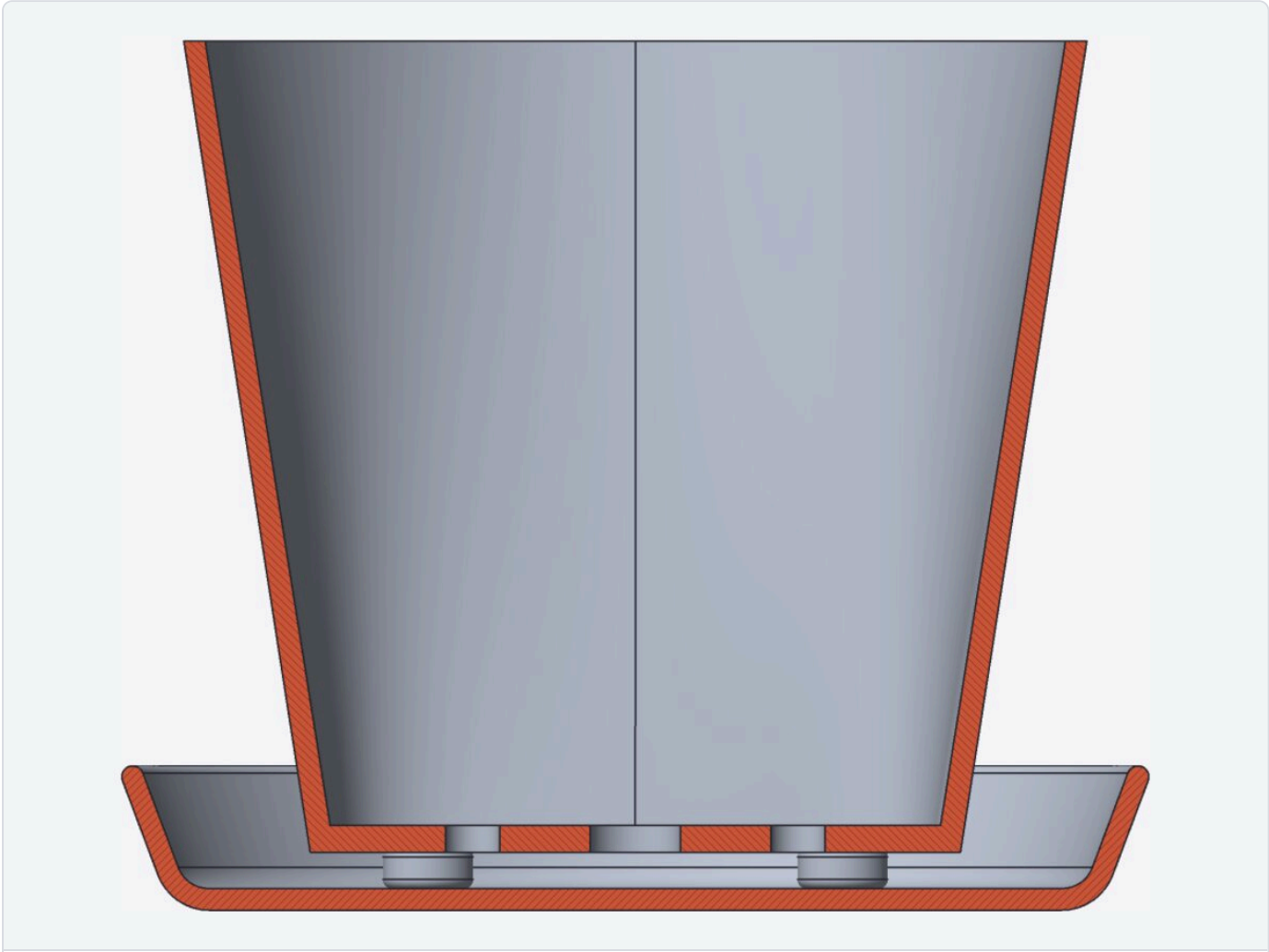
```
OPEN      cad tests/models/everyday/planter/planter.mcad
HEADLESS  cad render tests/models/everyday/planter/planter.mcad -o bild.png --views all
```

18 Plant pot with saucer – fit

Assembly

Part	Module	Bounding box (W × D × H) mm	Volume cm³	Mass PLA g
saucer	saucer	113.7 × 113.7 × 16.0	31.06	38.5
pot	pot	100.0 × 100.0 × 94.0	76.11	94.4

Pair (measure --clearance)	Distance	Meaning
pot ↔ saucer	0.000 mm	Feet stand on the saucer base
measure --interference	no collision (tolerance 0.001 mm³)	



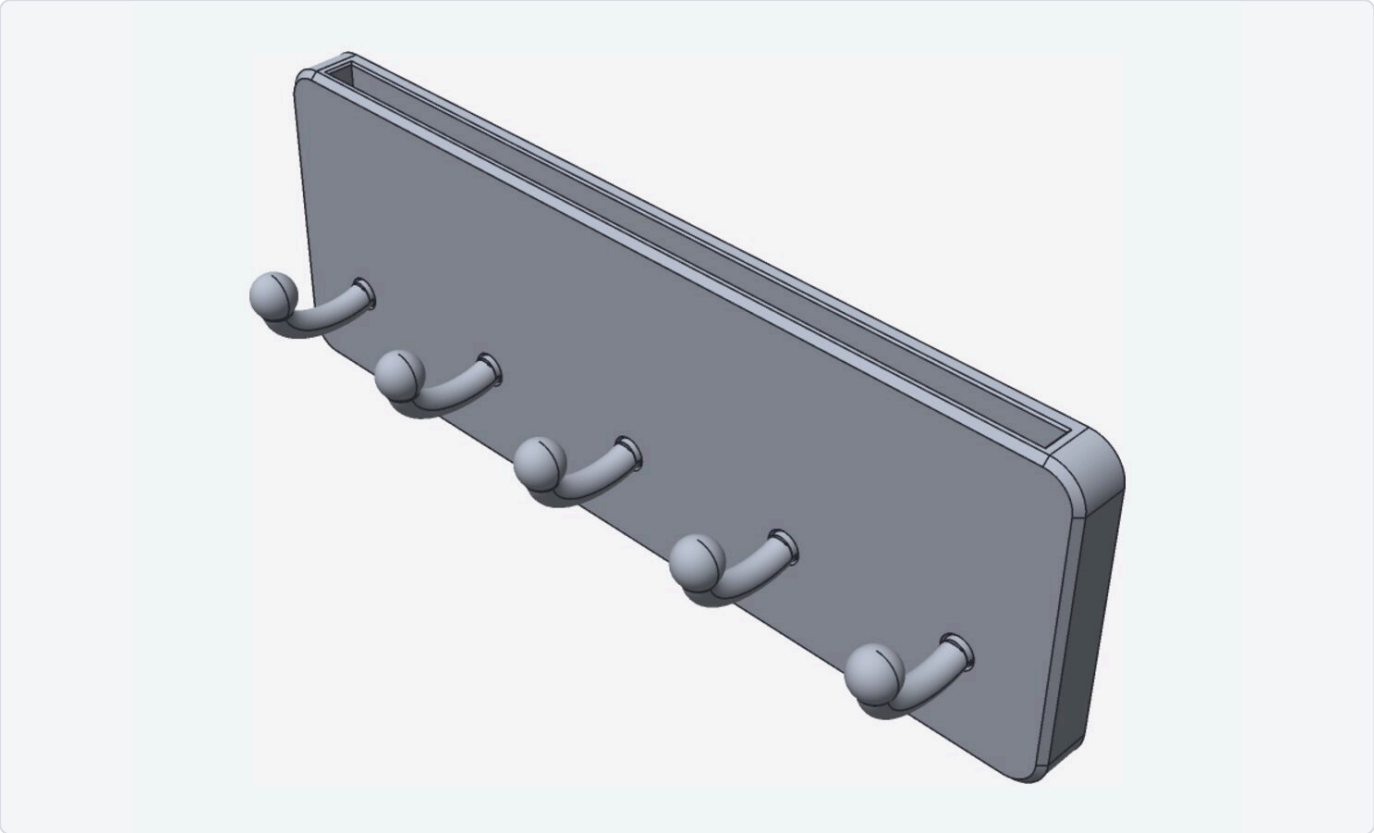
Section y = 0, view from the front: wall of the loft, feet and drainage holes, pot in the saucer.

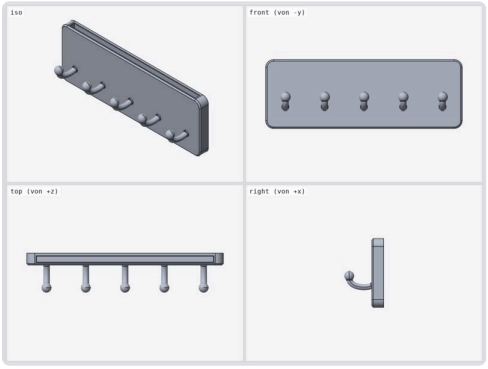
```
CHECK  cad measure tests/models/everyday/planter/planter.mcad --interference --clearance pot
       saucer
```

19 Key rack

Assembly

Board 200 × 70 × 12 mm with 5 glued-in hooks. It has corners filleted via zones and a chamfered front edge, two keyhole hangers at the back and a letter slot on top. The hooks are a sweep along a spline path with a ball knob; their peg Ø 6 mm sits in a hole Ø 6.3 mm (glue gap). The hooks are parts h[0] ... h[4] from a for loop, the dimensions are in dimensions.mcad.





Views iso, front, top, right (same scale)

Bounding box (W × D × H)	200.0 × 40.0 × 70.0 mm
Volume	158.1 cm³
Mass PLA (1.24 g/cm³)	196.0 g
Parts	6: `board`, `h[0]`, `h[1]`, `h[2]`, `h[3]`, `h[4]`
Colours / filaments	single colour (filament slot 1)
Kernel time (no cache)	3.41 s (68 nodes)

PARAMETERS (DEFAULT VALUES)

File `board_l = 200` `board_h = 70` `board_t = 12` `count = 5` `hook_z = 22` `peg_d = 6` `peg_l = 8` `bore_d = 6.3`

LANGUAGE FEATURES

grid · indexed grid planes · anchor · anchor with dir · named instances · include · use · function · list comprehension · for · spline · sweep · fillet zones · chamfer zones · zone · difference(fillet/chamfer) · union(fillet) · box(a, b) · cylinder(a, b, r) · sphere

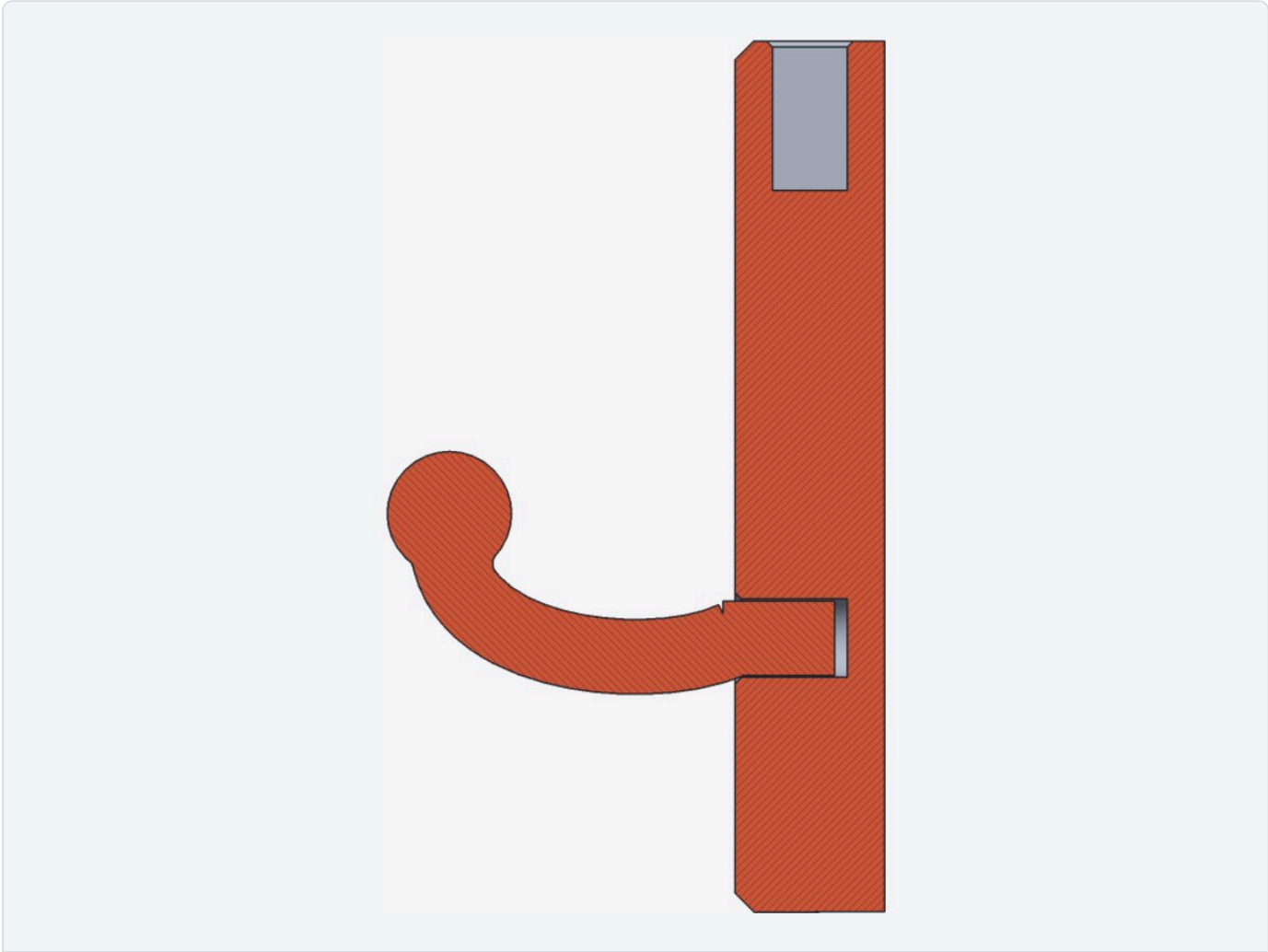
```
OPEN      cad tests/models/everyday/key_rack/key_rack.mcad
HEADLESS  cad render tests/models/everyday/key_rack/key_rack.mcad -o bild.png --views all
```

19 Key rack – fit

Assembly

Part	Module	Bounding box (W × D × H) mm	Volume cm³	Mass PLA g
board	board	200.0 × 12.0 × 70.0	150.72	186.9
h[0]	hook	10.0 × 36.0 × 19.5	1.47	1.8
h[1]	hook	10.0 × 36.0 × 19.5	1.47	1.8
h[2]	hook	10.0 × 36.0 × 19.5	1.47	1.8
h[3]	hook	10.0 × 36.0 × 19.5	1.47	1.8
h[4]	hook	10.0 × 36.0 × 19.5	1.47	1.8

Pair (measure --clearance)	Distance	Meaning
board ↔ h[0]	0.081 mm	Peg in the hole (glue gap)
measure --interference	no collision (tolerance 0.001 mm³)	



Section x = 20 (through hook h[0]), view from +x: peg in the hole, chamfer of the front edge, letter slot on top.

```
CHECK  cad measure tests/models/everyday/key_rack/key_rack.mcad --interference --clearance board
       h[0]
```